

Racial Gaps in Friendship Among High-Achieving Students: A Structural Account

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Abstract

High-achieving Black students have fewer friends than high-achieving White students. This pattern is often attributed to race-specific cultural norms—most notably, the "acting White" hypothesis—which posit that academically successful Black students face distinctive social costs from their own racial group. Using nationally representative friendship network data from the Add Health study, we revisit this interpretation. We show that the gap can be largely accounted for by two race-neutral forces: universal patterns of homophily—students of all races prefer friends who share their race and achievement level—and the demographic structure of schools, which leaves high-achieving Black students with a much smaller pool of same-race, high-achieving peers. Counterfactual exercises that equalize the demographic composition of schools eliminate most of the friendship gap among high achievers, without invoking any group-specific cultural attitudes toward achievement. Combining these estimates with existing causal estimates of the economic returns to friendship suggests that differences in adolescent social networks may have meaningful downstream consequences for racial inequality in adulthood.

friendship formation, homophily, racial friendship gap, racial earnings gap
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1 Introduction

Having friends in school shapes adolescents' academic trajectories, socioemotional development, and long-term economic prospects (Calvó-Armengol, Patacchini, and Zenou 2009; Carrell, Hoekstra, and Kuka 2018; Christakis and Fowler 2007; Coleman 1988; Nakajima 2007). Recent evidence shows that cross-class friendships can promote intergenerational mobility (Chetty et al. 2022a, 2022b), underscoring that social ties are not just personally enriching but economically consequential. Yet access to these ties is far from equal. Structural segregation by race and class limits many minority students' exposure to peers from higher socioeconomic or academic backgrounds (Ananat 2011; Cutler and Glaeser 1997; Massey 1990; Orfield and Frankenberg 2014), and even within integrated schools, strong tendencies toward same-race friendship formation—homophily—persist (McPherson, Smith-Lovin, and Cook 2001; Mele 2020; Moody 2001; Wimmer and Lewis 2010).

A particularly striking pattern emerges at the top of the achievement distribution: high-achieving Black students have substantially fewer friends than their White counterparts. The existing explanation for this deficit is the “acting White” hypothesis (Austen-Smith and Fryer Jr 2005; Fordham and Ogbu 1986; Fryer Jr and Torelli 2010), which holds that academically successful minority students face social sanctions from their own racial group for pursuing excellence. This account has been questioned on several grounds. Ainsworth-Darnell and Downey (1998) find that Black students report more pro-school attitudes than White students, undermining the premise of an oppositional peer culture. Tyson, Darity, and Castellino (2005) argue that achievement-related stigma reflects tracking structures rather than cultural opposition to achievement (see also Carter 2005; Mickelson 1990). Cross-national evidence raises further doubts. In Brazil, where friendship networks are more racially integrated and homophily is weaker, Portella, Kirschbaum, and Menezes-Filho (2022) find that non-White students do not face a trade-off between academic performance and peer acceptance. This suggests that the acting White pattern may be contingent on the degree of racial segregation in social networks rather than reflecting a cultural norm among minority groups. However, despite these challenges, the friendship deficit among high-achieving Black students remains largely unexplained, and the

acting White hypothesis continues to serve as a prominent account for the friendship gap.

We offer a fundamentally different account. Using social network data from the National Longitudinal Study of Adolescent to Adult Health (Add Health), a nationally representative study of U.S. adolescents, we show that the friendship gap among high achievers is largely accounted for by demographic structure, without invoking group-specific peer sanctions. Three empirical findings underpin this conclusion. First, friendship formation is governed by both racial and achievement homophily—students prefer friends of the same race and similar GPA. Second, the sensitivity of friendship to achievement differences is indistinguishable between Black–Black and White–White pairs, inconsistent with the prediction that Black students are penalized for high achievement. Third, achievement matters less in cross-racial friendships, implying that such ties cannot offset the scarcity of same-race high-achieving peers.

Counterfactual simulations that equalize the racial composition and GPA distribution of schools eliminate over 90% of the friendship gap among high-achieving students. That is, the combination of universal social preferences—the tendency to befriend those of the same race and similar achievement—with the demographic structure of schools could account for the majority of the gap without invoking any race-specific cultural norms. The key constraint is demographic: high-achieving Black students are a numerical minority with few same-race peers at similar achievement levels. We further show that this structural deficit carries economic consequences, potentially accounting for 14–30% of the Black–White earnings gap among high achievers.

2 Results

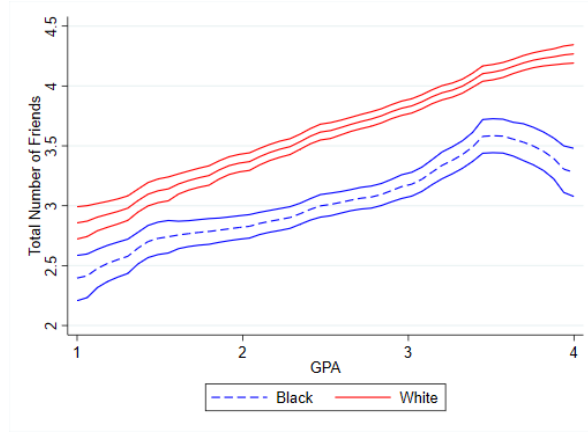
2.1 The friendship gap widens with achievement.

The central empirical puzzle motivating our analysis is visible in the raw data. Figure 1, panel (a), plots the total number of friends across the GPA distribution, separately for Black and White students. On average, Black and White students have 3.04 and 3.80 friends, respectively.¹ Both groups gain friends as GPA rises, but the rate of increase is far steeper for White students.

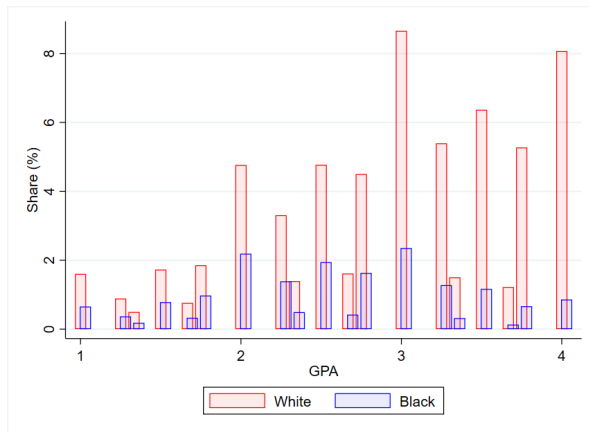
¹ The number of friends is based on friendship nominations. Because each student could nominate up to five male and five female friends, the measure may be influenced by the nomination cap. White students hit the cap more frequently (58%) than Black students (47%), suggesting that the true friendship gap may be larger than observed. Our estimates are therefore conservative.

Among high-achieving students (GPA above 3.75), the gap reaches almost one friend, which is about 30% of the total. This widening gap is the phenomenon we seek to explain.

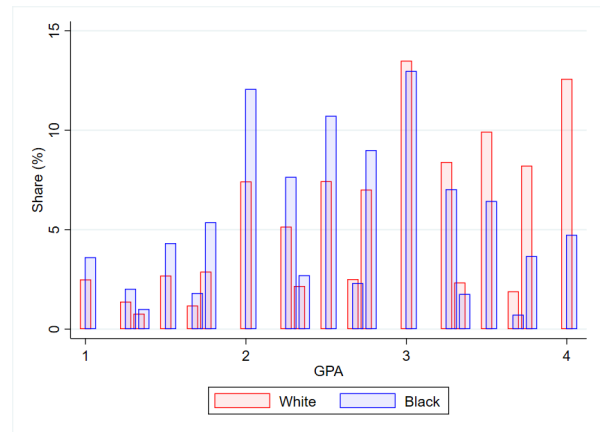
Fig. 1. Friendship Patterns and GPA Distributions by Race



(a) Number of Friends by GPA



(b) GPA Distribution by Race: Original Shares



(c) GPA Distribution by Race: Equal Shares

Notes. Panel (a) shows locally smoothed averages of the number of friends across the GPA distribution, separately for Black and White students. Panels (b) and (c) illustrate the distributions of average GPA for Black and White students. The red bars represent the density of the average GPA of White students, and the blue bars represent the density of the average GPA of Black students. Panel (b) presents the distribution using the original shares, while panel (c) shows the distribution after equalizing the shares of Black and White students.

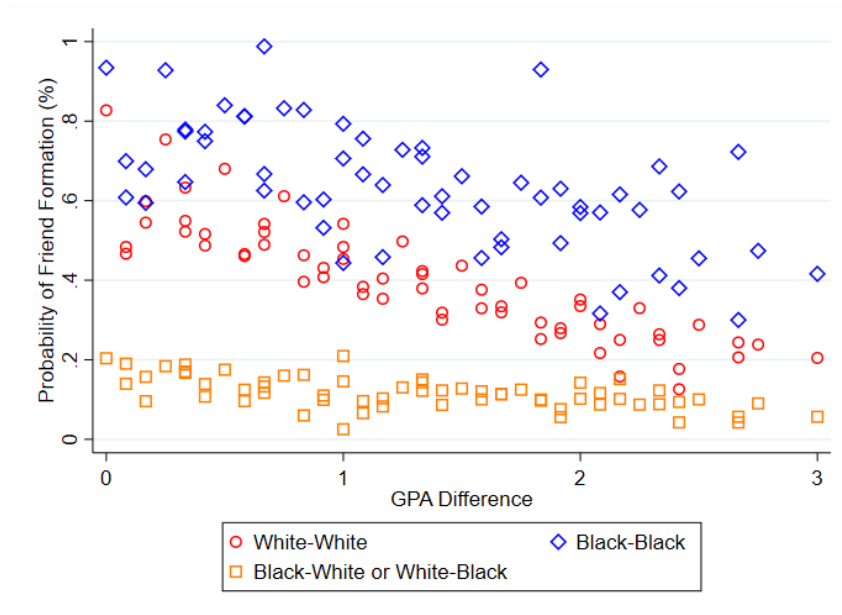
The structural origin of this pattern becomes apparent when we examine school composition, which prior work has identified as a key determinant of cross-race contact opportunities (Hallinan and Williams 1989; Joyner and Kao 2000; Quillian and Campbell 2003). Figure 1 shows the GPA distributions of Black and White students under observed racial shares (b) and after equalizing the proportion of each group (c). White students dominate most schools numerically and are disproportionately concentrated in the upper GPA range; Black students, by contrast, are concentrated in the middle range. Even after equalizing group sizes, the Black GPA distribution

remains left-shifted relative to the White distribution.² For a high-achieving Black student in a typical U.S. school, the pool of potential same-race, high-achieving friends is doubly constrained: by the small number of Black students in the school and by the low proportion of high achievers among them.

2.2 Homophily operates similarly across races.

To move beyond description, we estimate a pairwise regression model of friendship formation as a function of race-pair indicators and GPA differences. Figure 2 displays the key result graphically: the probability of friendship declines with GPA distance for all race pairs, but the decline is steeper for within-race pairs (White–White and Black–Black) than for cross-race pairs (Black–White and White–Black).

Fig. 2. Probability of Forming Friendships by GPA Difference and Race



Notes. The figure illustrates the probability of forming friendships based on the GPA difference between pairs, depicted separately for racial combinations. Each potential dyad of students in each school is treated as an observation.

Table 1 formalizes this pattern using the dyadic regression framework outlined in Eq. 1 in *Materials and Methods*. Three findings emerge from this analysis. First, a one-unit increase in the GPA gap between two White students reduces the probability of friendship by 0.189 percentage points. Given a mean friendship probability of 0.53%, this represents a 36% decrease relative to the mean. The corresponding interaction for Black–Black pairs ($dGPA \times BB$) is

² The pattern remains consistent even in predominantly Black schools (SI Appendix A.4).

statistically insignificant, meaning that Black and White students exhibit the same sensitivity to achievement differences within their racial group. If high-achieving Black students faced additional social sanctions for academic success—as the acting White hypothesis predicts—we would expect a stronger negative effect of GPA distance among Black pairs. The absence of such a differential suggests that the social costs high-achieving Black students face arise not from race-specific peer sanctioning but from the demographic constraints documented above.

Second, cross-racial friendships are substantially less common and less sensitive to GPA differences. The baseline probability of friendship is significantly lower for cross-race pairs (BW and WB), and the interactions for Black–White pairs ($dGPA \times BW$ and $dGPA \times WB$) are positive and significant, indicating that achievement similarity matters less when students cross racial lines. As a result, cross-race ties do not fully offset the scarcity of same-race, high-achieving peers for Black students.

Third, these patterns survive the inclusion of individual sender and receiver fixed effects (column 5), which absorb all time-invariant characteristics including unobserved social skills, personality, and family background. They also hold across multiple robustness checks: alternative achievement measures, within-club comparisons, mutual friendship definitions, asymmetric GPA effects, semiparametric specifications, and inclusion of Hispanic students (SI Appendix, [B](#)).

One implication of these estimates is that Black and White students may face asymmetric social trade-offs from raising their GPA. In SI Appendix Fig. [A.2](#), we use the estimated coefficients to compute how expected friendship counts change across the GPA distribution by race. Because the White GPA distribution is concentrated at higher levels than the Black distribution, White students tend to gain friends as GPA rises, whereas Black students lose same-race friends, with cross-race friendships only partially compensating. Because this exercise relies on a simplified model, the magnitudes should be interpreted with caution; nonetheless, the racial pattern aligns with observed friendship counts.

Table 1: Friendship Formation by Race and Achievement

	Dependent variable : <i>i</i> nominating <i>j</i> as a friend (%)				
	(1)	(2)	(3)	(4)	(5)
<i>dGPA</i>	-0.189*** (0.019)	-0.164*** (0.015)	-0.216*** (0.025)	-0.189*** (0.022)	-0.259*** (0.028)
<i>dGPA</i> × <i>BB</i>			0.047 (0.044)	0.017 (0.045)	0.038 (0.049)
<i>dGPA</i> × <i>BW</i>			0.188*** (0.025)	0.158*** (0.022)	0.200*** (0.025)
<i>dGPA</i> × <i>WB</i>			0.182*** (0.024)	0.158*** (0.023)	0.193*** (0.026)
<i>BB</i>			0.168 (0.166)	0.250 (0.171)	
<i>BW</i>			-0.624*** (0.078)	-0.627*** (0.061)	
<i>WB</i>			-0.597*** (0.078)	-0.603*** (0.060)	
Constant	0.674*** (0.060)	0.694*** (0.097)	0.752*** (0.080)	0.862*** (0.129)	0.938*** (0.051)
Observations	36,183,256	36,183,256	36,183,256	36,183,256	36,183,256
Mean	0.513	0.513	0.513	0.513	0.513
Controls		X		X	X
Individual FE					X

Notes. The table presents estimation results from equation (1). Units of observation are directed dyads, based on the non-Hispanic Black and non-Hispanic White students who reported GPAs and individual characteristics (age, gender, and race). The dependent variable takes the value of 1 if student *i* nominates student *j* as a friend. *dGPA* indicates a difference in average GPA within a pair of students. *B* indicates a Black student, and *W* indicates a White student. *BW* is an indicator that the sender is a Black student and the receiver is a White student. “Mean” refers to the average probability of forming friendships. “Controls” indicate the controlling demographic variables including gender and age of senders and receivers, difference in age within pairs, and school fixed effects. “Individual FE” indicates that individual fixed effects of senders and receivers are controlled for. Standard errors are clustered at the school level. *** $p < .01$, ** $p < .05$, * $p < .1$

2.3 Demographic structure accounts for the gap.

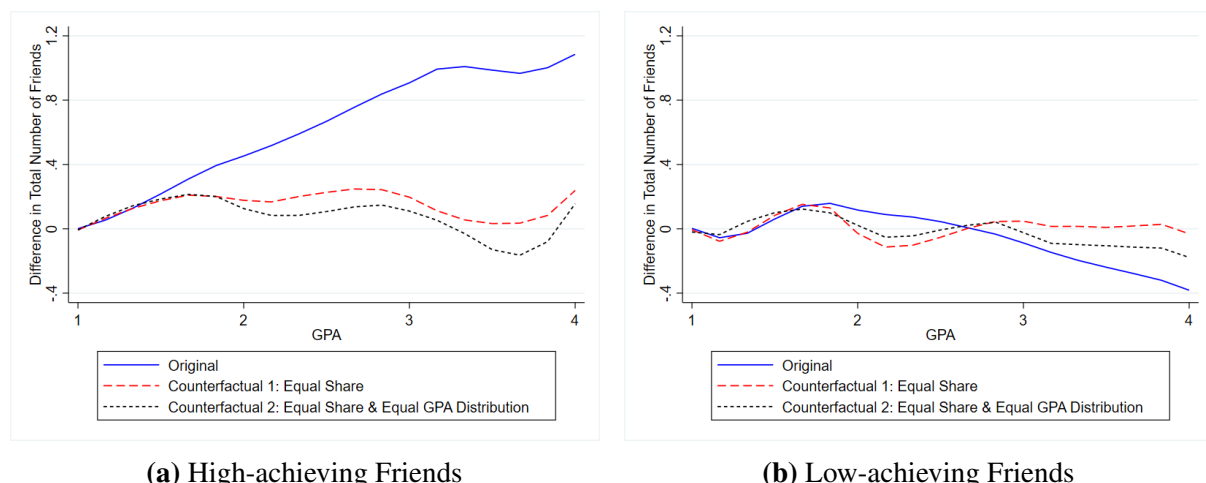
The regression results establish that homophily operates similarly across races. But can structural factors alone—group size and achievement distribution—quantitatively account for the friendship gap? To answer this, we build a simulation model that uses the estimated friendship formation coefficients and observed student distributions within each school to compute expected friendship counts by race and GPA (see Eq.2 in *Materials and Methods*).

The model closely reproduces the observed racial gap in friendship counts. Simulated and actual differences between Black and White students’ friend counts differ by less than 0.2

across all GPA levels (SI Appendix Fig. A.5 and SI Table A.7). This close fit confirms that the simulation captures the key forces driving the racial friendship gap and can serve as a reliable baseline for the counterfactual exercises that follow.

We then perform two counterfactual exercises. First, we equalize the number of Black and White students in each school (*level effect*). Second, we additionally replace the Black GPA distribution with the White distribution (*composition effect*). To isolate the widening of the gap across the achievement distribution, we normalize the friendship gap to zero at the lowest GPA level. Figure 3 presents the results for high-achieving friends (a) and low-achieving friends (b).

Fig. 3. Counterfactual Decomposition of the Racial Friendship Gap



Notes. The figure illustrates the difference in the number of expected friends between Black and White students (White minus Black), standardized to 0 at the lowest GPA level. The solid blue line depicts the original difference. The dashed red line represents the difference after equalizing the proportion of Black and White students. The short-dashed black line represents the difference after equalizing both shares and GPA distributions.

For high-achieving friends, equalizing racial shares reduces the gap from 1.05 to 0.16 among students with a GPA of 3–4, an 85% reduction. Further equalizing the GPA distribution brings the gap down to 0.07. Together, these two structural adjustments explain 93% of the friendship gap among high achievers. The dominant factor is group size: because Black students are typically a numerical minority, equalizing racial shares removes most of the difference in representation among high achievers (see Figure 1 panel (b)), leaving high-achieving Black students with fewer potential same-race, high-achieving friends.

For low-achieving friends, the pattern reverses. High-achieving Black students actually have slightly *more* low-achieving friends than White students at the same GPA, reflecting their greater

proximity to the mean of the Black GPA distribution. This asymmetry further confirms the structural mechanism: the friendship gap is not about social rejection of achievement, but about the demographic composition of the available peer pool.

These findings are robust to alternative coefficient sets. Using estimates from all schools or from schools with 23–40% Black enrollment yields nearly identical results (SI Appendix Fig. A.9). SI Appendix Table A.8 provides numerical values for each counterfactual step. The structural mechanism also generates a clear cross-school prediction: the friendship gap should widen with achievement primarily in schools where Black students are a numerical minority, because it is in those settings that the pool of same-race, high-achieving peers is most constrained. Consistent with this prediction, the gap is concentrated in schools with low Black enrollment and is absent in majority-Black schools (SI Appendix A.4, Fig. A.3).

Importantly, nothing in our analysis rules out the existence of peer sanctioning for academic achievement. What the results demonstrate is that such sanctions are not necessary to account for the observed friendship gap; demographic structure alone can generate it.

2.4 The friendship gap has lasting economic consequences.

Adolescent friendships are not only socially meaningful—they provide information, motivation, and access to opportunities that shape long-term outcomes (Calvó-Armengol and Jackson 2004; Chetty et al. 2022b; Coleman 1988; Crosnoe, Cavanagh, and Elder 2003; Jackson et al. 2008; Stanton-Salazar and Dornbusch 1995). If high-achieving Black students systematically have fewer friends, the consequences may extend well beyond school.

To illustrate the potential magnitude of these consequences, we combine our counterfactual estimates of the friendship deficit with instrumental variable (IV) estimates of the return to friendship (see Eq. 4 in *Materials and Methods*). Following the identification strategy of Lleras-Muney et al. (2025), we estimate that one additional friend increases annual earnings, with no significant difference between Black and White students. Our point estimates are consistent with the 6.5–13.7% bounds reported in their study, though imprecisely estimated due to limited statistical power in race-specific specifications (SI Appendix Table A.11). We therefore adopt their bounds, rather than our own point estimates, for the back-of-the-envelope calculations that

follow.

Our counterfactual analysis implies that equalizing school composition would increase high-achieving Black students' friendship count by 0.87 (0.98 high-achieving friends gained, 0.11 low-achieving friends lost). Applying the estimated returns, this translates to a 5.7–11.9% increase in annual earnings. In our data, high-achieving Black workers earn 28.7% less than their White counterparts (average earnings: \$37,300).³ The estimated earnings increase of 5.7–11.9% would reduce this gap to 20.2–24.6%—a reduction of 14–30%. These calculations should be interpreted as illustrative rather than definitive, as they combine our estimates of friendship gaps with causal estimates from prior work.

While these earnings estimates are necessarily approximate, they suggest that the demographic composition of a student's peer group could generate economically meaningful inequality through a purely structural channel, even in the absence of differential treatment or discrimination.

3 Discussion

We have shown that the friendship deficit of high-achieving minority students is structural in origin and consequential for long-run inequality. Two demographic facts—that Black students are a numerical minority in most schools, and that fewer of them are high achievers—combine with universal patterns of racial and achievement homophily to produce a friendship gap that widens with academic achievement. Counterfactual simulations demonstrate that adjusting the demographic composition of schools eliminates most of this gap, suggesting that the deficit could arise primarily from the composition of available peers.

Our findings contribute to the debate surrounding the “acting White” hypothesis by offering network-level evidence that complements earlier critiques based on attitudinal surveys (Ainsworth-Darnell and Downey 1998) and ethnographic studies (Carter 2005; Tyson, Darity, and Castellino 2005). The hypothesis predicts that friendship formation should be more sensitive to achievement differences among Black pairs than among White pairs. We find no evidence of such a differential: achievement homophily operates with similar strength across racial groups.

³ After controlling for demographics, social skills, cognitive skills, and school and grade fixed effects, the gap narrows to 20.3%.

However, because this universal tendency interacts with the demographic structure of most schools—where high-achieving Black students constitute a small share of an already small group—the same homophily that benefits high-achieving White students by connecting them to a large pool of same-race, high-achieving peers leaves Black students with far fewer such connections.

The economic implications are substantial. Our estimates suggest that the structural friendship deficit could account for 14–30% of the Black–White earnings gap among high-achieving young adults. This finding adds to a growing body of evidence linking social network structure to economic mobility (Chetty et al. [2022a](#), [2022b](#)) and highlights the demographic composition of peer networks as an underappreciated source of racial inequality in social capital.

These results also carry implications for school integration policy. Intergroup contact theory (Allport [1954](#); Pettigrew and Tropp [2006](#)) predicts that increased exposure to outgroup members reduces prejudice, and school desegregation efforts have long been motivated by this logic (Ananat [2011](#); Angrist and Lang [2004](#)). However, our analysis suggests that policies increasing racial diversity at the school level may be insufficient if they do not also foster within-school connections across racial and achievement lines. Transporting minority students to majority-White schools may not reduce social segregation, because racial homophily limits cross-race friendship formation (Moody [2001](#)). High-achieving Black students in such settings may still lack same-race high-achieving peers—a deficit that cross-race ties cannot fully offset. Effective interventions would need to address both the demographic composition of schools and the conditions under which cross-group friendships form (Small [2009](#)).

More broadly, our findings illustrate how persistent racial inequality in social networks can emerge from the interaction of universal social preferences—the tendency to befriend similar others (McFarland et al. [2014](#); McPherson, Smith-Lovin, and Cook [2001](#))—with demographic structure (Bruch and Mare [2006](#))—the underrepresentation of Black students, particularly among high achievers, in most U.S. schools. Recognizing the role of demography alongside culture opens new avenues for understanding and addressing racial disparities in social capital.

4 Materials and Methods

4.1 Data.

We use data from the National Longitudinal Study of Adolescent to Adult Health (Add Health), a nationally representative, school-based study of U.S. adolescents. The in-school survey was administered to approximately 90,000 students in grades 7–12 across 142 schools during the 1994–1995 academic year. Each student could nominate up to five male and five female friends, and we observe the identity of each nominated friend within the same school. Our measure of friendship is based on the number of nominations received.⁴

Our primary sample consists of non-Hispanic Black and non-Hispanic White students who reported GPA and demographic information (age, gender, race). We construct directed dyads: for each pair of students (i, j) in the same school, we record whether i nominated j as a friend, yielding over 14 million observations. Academic achievement is measured as the average of self-reported grades in English, math, history, and science (1–4 scale). The mean probability of forming a friendship in any given dyad is 0.53%. Summary statistics are reported in SI Appendix Tables A.1 and A.2.

For the earnings analysis, we link adolescent friendship data to adult outcomes using Add Health Wave 4 (2007–2008), when respondents were aged 24–34. The outcome is total earnings from wages or salary in the previous year. Average earnings in the sample are approximately \$37,300; high-achieving Black workers earn 28.7% less than their White counterparts.

The Add Health data are available through a restricted-use contract from the Carolina Population Center at the University of North Carolina at Chapel Hill (<https://addhealth.cpc.unc.edu/>).

⁴ As noted in Section 2.1, 47% of Black students listed five friends of at least one gender, compared to 58% of White students. If this ceiling reflects binding nomination constraints, the true friendship gap may be larger than what we observe, because White students—who more frequently hit the cap—are more likely to have understated friendship counts. Our estimates of the racial friendship gap are therefore conservative.

4.2 Pairwise regression.

We estimate the following model on directed dyads:

$$G_{ij} = \alpha_0 + \beta_0 d_{ij} + \beta_{bb} d_{ij} \times B_i B_j + \beta_{bw} d_{ij} \times B_i W_j + \beta_{wb} d_{ij} \times W_i B_j \\ + \alpha_{bb} B_i B_j + \alpha_{bw} B_i W_j + \alpha_{wb} W_i B_j + \mathbf{X}' \boldsymbol{\delta} + \varepsilon_{ij}, \quad (1)$$

where G_{ij} equals 1 if student i nominated student j as a friend, d_{ij} is the absolute GPA difference, and race-pair indicators capture deviations from the White–White baseline. $BB = 1$ when both students are Black; $B_i W_j = 1$ when the sender is Black and the receiver is White; $W_i B_j$ for the reverse. The coefficient β_0 captures the effect of GPA distance on friendship probability for White–White pairs, while β_{BB} , β_{BW} , and β_{WB} capture how this effect differs for other race pairs. $\mathbf{X}$ includes gender and age of both students, age difference, and school fixed effects. We also estimate specifications with sender and receiver fixed effects. Standard errors are clustered at the school level.

4.3 Simulation and counterfactual decomposition.

Using school-specific estimated coefficients and observed GPA distributions, we compute the expected number of friends for a Black student with GPA x :⁵

$$\tilde{n}_x^B = \sum_{s \in E} N_B^s \times \int_{lb}^{ub} \left[(\hat{\alpha}_0^s + \hat{\alpha}_{BB}^s) + (\hat{\beta}_0^s + \hat{\beta}_{BB}^s) |x - z| \right] f_B^s dz \\ + \sum_{s \in E} N_W^s \times \int_{lb}^{ub} \left[(\hat{\alpha}_0^s + \hat{\alpha}_{BW}^s) + (\hat{\beta}_0^s + \hat{\beta}_{BW}^s) |x - z| \right] f_W^s dz, \quad (2)$$

where N_B^s and N_W^s are the number of Black and White students in school s , f_B^s and f_W^s are GPA distributions, and $\hat{\alpha}^s$ and $\hat{\beta}^s$ are school-specific parameters. E is the set of schools with sufficient variation to estimate all race-pair coefficients. For high-achieving friends, $lb = 3$, $ub = 4$; for low-achieving friends, $lb = 1$, $ub = 3$.

We normalize the friendship gap to zero at the lowest GPA level because our quantity of interest is the widening of the gap with achievement—the phenomenon that motivates the acting

⁵ The corresponding expression for White students is defined analogously (SI Appendix, Eq. A.2)

White hypothesis—rather than the overall level difference in friendship counts. Without this normalization, equalizing racial shares produces a mechanical level shift: homophily parameters estimated from a small Black population may overstate absolute same-race friendship counts when group size is increased, because the linear model does not incorporate friendship capacity constraints. The normalization removes this level component and focuses the comparison on how the gap changes across the GPA distribution. Results without normalization are reported in SI Appendix Fig. A.7.

$$d(x) = [\tilde{n}^W(x) - \tilde{n}^B(x)] - [\tilde{n}^W(x=1) - \tilde{n}^B(x=1)]. \quad (3)$$

Two counterfactual exercises modify this gap: (i) equalizing the number of Black and White students in each school (*level effect*) and (ii) additionally replacing the Black GPA distribution with the White distribution (*composition effect*). For details, see SI Appendix A.5.

4.4 Earnings analysis.

We estimate the causal return to friendships using an IV approach following Lleras-Muney et al. (2025):

$$\ln(\text{earnings})_i = \gamma_0 + \gamma_f \text{NumFriend}_i + \gamma_b (\text{Black}_i \times \text{NumFriend}_i) + \iota X_{ijg} + \delta_j + \eta_g + e_i. \quad (4)$$

The instrument is the average absolute age difference between a student and peers in the same school and grade. Age similarity promotes friendship formation (McPherson, Smith-Lovin, and Cook 2001) while being plausibly exogenous to earnings. Placebo tests by Lleras-Muney et al. (2025) confirm that age distance is unrelated to predetermined characteristics linked to earnings, including parental education and birth weight. We additionally construct race-specific age distance measures to account for racial homophily in friendship formation. Controls include age, sex, race, social skills (self-reported extroversion), cognitive skills (picture vocabulary test), and school and grade fixed effects. SI Appendix Table A.10 reports first-stage results; SI Appendix Table A.11 reports OLS and 2SLS estimates.

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ONLINE APPENDIX

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A Additional Results

A.1 Summary Statistics

Table A.1: Individual-Level Summary Statistics

	White	Black	All
GPA	2.9	2.6	2.8
Female (%)	51	55	52
Age	15	15	15
Number of Friends	3.80	3.04	3.64
High Performing Friends ($GPA \geq 3$)	2.38	1.40	2.17
Low Performing Friends ($GPA < 3$)	1.42	1.64	1.47
Share (%)	78.9	21.1	100
Observations	43,109	11,530	54,639

Notes. The table presents individual-level summary statistics for White, Black, and all students. It reports the average number of friends, broken down into two categories: high-performing friends ($GPA \geq 3.0$) and low-performing friends ($GPA < 3.0$).

Table A.2: Pairwise-Level Summary Statistics

	White→White	Black→Black	White→Black	Black→White	All
Friend (%)	0.567	0.784	0.125	0.103	0.513
GPA diff.	0.9	0.8	0.9	0.9	0.9
Share (%)	73.29	9.91	8.40	8.40	100
Observations	26,517,518	3,584,014	3,040,862	3,040,862	36,183,256

Notes. “Friend (%)” measures the proportion of formed friendships out of all possible student pairs. If student i nominates student j as a friend but j does not reciprocate, only the i - j pair is counted as a friendship, not the j - i pair. “GPA diff.” refers to the average GPA gap between paired students.

A.2 Calculating Probability of Forming Friends

The following illustration outlines the methodology employed to generate Figure 2. Consider a school with a total of 15 students. Within this group, there are two high-achieving Black students, three low-achieving Black students, five high-achieving White students, and five low-achieving White students.

Table A.3: Example with Pairs of 15 Students

i and j	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
	B_H	B_H	B_L	B_L	B_L	W_H	W_H	W_H	W_H	W_H	W_L	W_L	W_L	W_L	W_L
1 B_H	-	1	1	1	0	0	0	0	0	1	0	0	0	0	1
2 B_H	1	-	0	1	1	0	1	0	0	0	0	0	0	0	0
3 B_L	1	0	-	1	1	0	0	0	0	1	0	0	0	0	0
4 B_L	0	0	1	-	0	0	0	0	0	0	0	0	0	1	0
5 B_L	1	1	1	1	-	0	0	0	0	1	1	0	1	0	0
6 W_H	0	0	0	0	0	-	1	1	1	0	1	0	0	0	1
7 W_H	0	0	0	0	0	1	-	1	0	1	0	0	0	0	0
8 W_H	0	0	0	0	0	1	1	-	0	0	1	0	0	0	1
9 W_H	0	0	0	0	0	1	0	0	-	1	0	0	0	0	0
10 W_H	1	1	1	0	1	0	1	1	0	-	1	0	0	0	1
11 W_L	0	0	0	0	1	1	0	1	1	1	-	1	0	1	0
12 W_L	0	0	0	0	0	0	0	0	0	0	0	-	1	0	0
13 W_L	0	0	0	0	0	0	0	0	0	0	1	0	-	1	0
14 W_L	0	0	0	1	0	0	0	0	0	0	0	0	1	-	1
15 W_L	1	0	0	0	0	1	0	1	0	1	1	1	0	1	-

Notes. This is a simple example with Black and White students with either high ability or low ability. Each row indicates a student who lists another student and each column indicates the student who are listed.

In Appendix Table A.3, we present the friendship connections among the 15 students. Each cell indicates whether student i in the row directs student j in the column as a friend. The labels in the table represent the race and achievement of the students, denoted as R_a , where R represents the race and a represents the achievement of the student. For instance, B_H in the first row (student $i = 1$) represents a high-achieving Black student. In the table, we can observe that student 2, who is a high achiever, directs student 7, who is a high achiever.

In Appendix Table A.4, we calculate the probabilities of forming friendships based on the data in Appendix Table A.3. For instance, consider the case of a high-achieving White student as the friendship sender and a low-achieving Black student as the friendship receiver. In this scenario, there are 15 potential pairs (5 high-achieving White students and 3 low-achieving

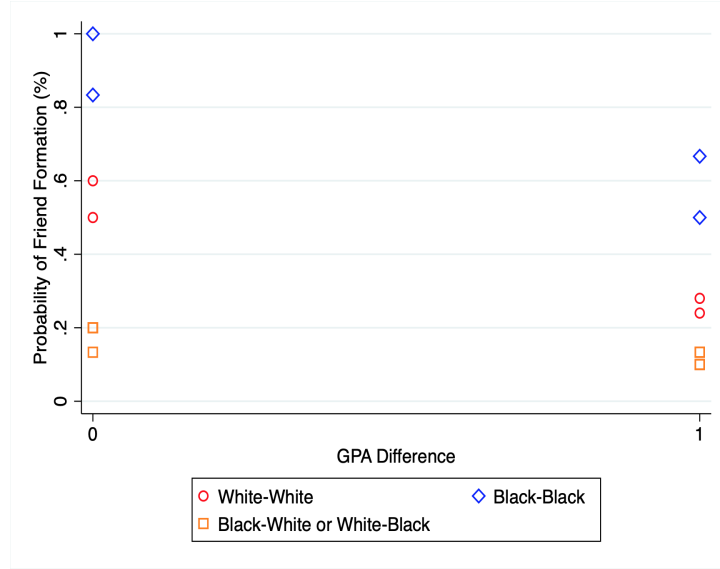
Table A.4: Probability of Forming Friendships

	B_H	B_L	W_H	W_L
B_H	100%	66.7%	20%	10%
B_L	50%	83.3%	13.3%	20%
W_H	20%	13.3%	60%	24%
W_L	10%	13.3%	28%	50%

Notes. We calculate probabilities of forming friendship for each possible race and achievement based on Appendix Table A.3.

Black students) that can form a friendship. However, only 2 actual friendships are formed out of these potential pairs. Therefore, the probability of a high-achieving White student directing a low-achieving Black student as a friend is $\frac{2}{15} = 13.3\%$.

Fig. A.1. Probability of Forming Friendships by GPA Difference and Race



Notes. The figure illustrates average probability of friendship formation by GPA difference and race pairs based on the Appendix Table A.3. In this figure, GPA difference is 0 when two students are both high type or low type. GPA difference is 1 when two students are different achievement type (high-low or low-high types).

Using Appendix Table A.3, we plot the relationship between the GPA difference and the probability of friendship formation for each possible pair in Appendix Figure A.1. For instance, there are two cases where two Black students have identical GPAs: a high-achieving Black student directing another high-achieving Black student as a friend, and a low-achieving Black student directing another low-achieving Black student as a friend. In the figure, the likelihood of a high-achieving Black student directing another high-achieving Black student as a friend is 100%, while the likelihood of a low-achieving Black student directing another low-achieving

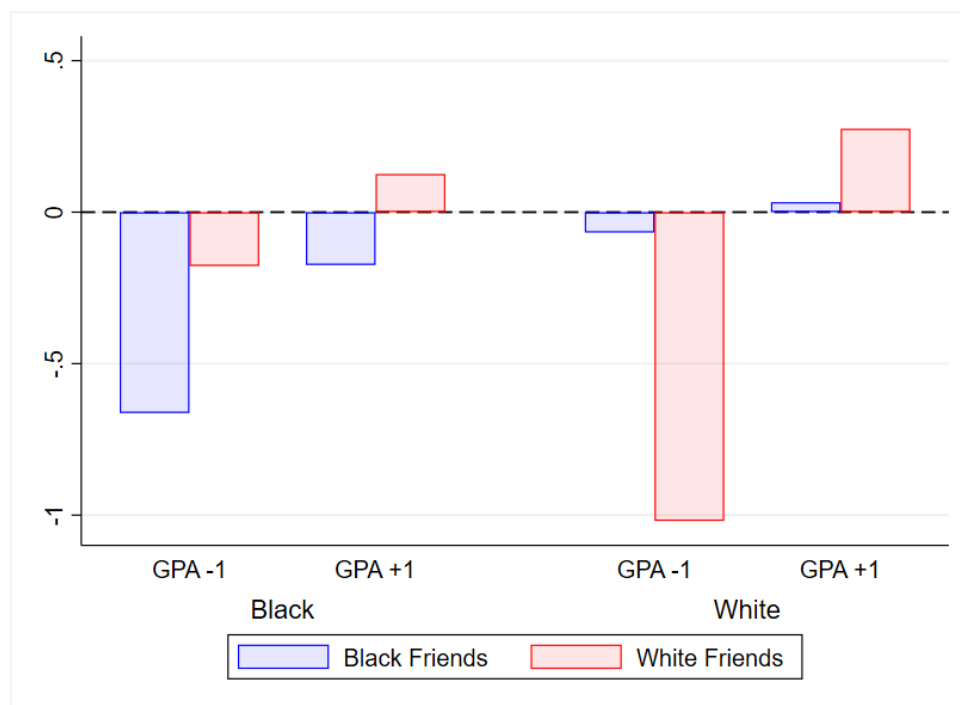
Black student as a friend is 83.3%.

Compared to these equally-achieving friend pairs, the likelihood of forming a friendship decreases when students have different GPAs. The likelihood of forming a friendship between Black students with different achievement is lower than that of the equally-achieving cases. Additionally, the likelihood of forming a friendship for same-race friendships is higher than that of different-race friendships for a given GPA difference.

A.3 Expected Friendship Changes with GPA Variation

One important implication of the regression results is that students may face different trade-offs when increasing their GPA. For example, a Black student with the average GPA of their group (2.6) might lose friends by moving away from the group's mean achievement, whereas a White student with the same GPA may not, due to differences in GPA distribution across racial groups. Using the expected number of friends from the regression results in column (4) of Table 1, we examine how friendship patterns vary across the GPA distribution by race. Figure A.2 shows the change in the expected number of friends between students with the average GPA for Black students and those with a GPA one point above or below that average. As GPA increases, White students tend to gain more friends, while Black students lose same-race friends. This loss is only partially offset by gains in cross-race friendships: overall, Black students lose 0.05 friends, while White students gain 0.31. These results should be interpreted with caution. Because they are based on a simplified model of expected friendships, they may not capture all aspects of friendship formation. For instance, the model does not account for the general popularity of high-achieving students, who may in reality gain more friends as their GPA rises. Nonetheless, as we show in the next section, the model closely replicates observed racial differences in friendship counts, making it valid for racial comparisons. Conversely, as GPA decreases, White students lose 0.25 more friends than Black students. While this is a descriptive exercise, it suggests that students may face different social incentives for academic achievement, shaped by homophily and the racial distribution of academic performance. It also helps explain why high-achieving minority students tend to have fewer friends than their majority peers.

Fig. A.2. Difference in the Number of Expected Friends Between Students with Average GPA and Students with Higher/Lower GPA



Notes. The figure illustrates the difference in the number of expected friendships between students with the average GPA of Black students and students with a GPA that is higher or lower by 1 than this average. The blue and red bars represent the differences in Black and White friends, respectively. The expected numbers are computed based on coefficients from equation (1).

A.4 Heterogeneity Across Black Shares in Schools

In this subsection, we explore heterogeneity in friendship formation across schools with different racial compositions. Racial composition may also modify the relationship between race and achievement as factors in friendship formation. We divide schools into four groups so each group has the same number of Black students: the first, second, third, and fourth groups include schools with Black student shares ranging from 0% to 22%, 23% to 40%, 41% to 74%, and 75% to 100%, respectively.

In Appendix Figure A.3, we first present the number of friends, categorized by race and the share of Black students. A few patterns are worth noting. High-achieving Black students in Black-minority schools (0%-40%) tend to have significantly fewer friends than their White peers, whereas this disparity is absent in schools in which the Black population is not a minority (41%-100%). Low-achieving Black students in Black-minority schools also exhibit differences in the number of friends compared to their White counterparts, although the difference is smaller than that observed among high-achieving students. Moreover, we present the GPA distribution of Black and White students by the proportion of Black students in Appendix Figure A.4. While White students tend to cluster above a GPA of 3 and Black students are concentrated between 2 and 3, Black-minority schools show more pronounced differences in the distribution of GPAs.^{A.1} In essence, the figure suggests that high-achieving Black students tend to have a smaller circle of friends compared to their White peers, especially in schools in which the proportion of Black students is smaller, particularly at higher GPAs.

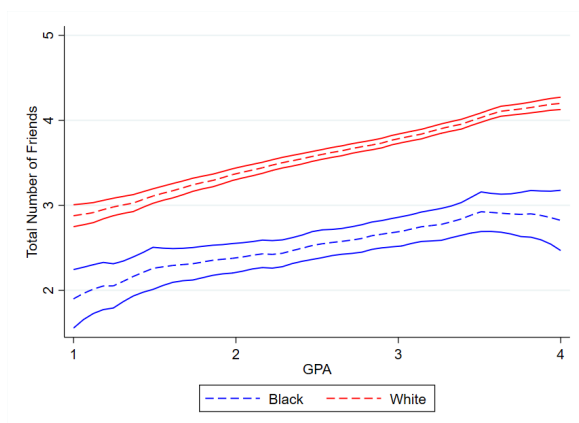
In Appendix Table A.5, we estimate equation (1) separately for students enrolled in schools with different shares of Black students. This analysis enables us to explore how friendship-formation patterns vary across shares of Black students. We present the estimation results starting with all schools in column (1), followed by schools categorized based on different shares of Black students in columns (2) to (5).

Across all school types, GPA differences are either similarly important or less pronounced for Black students than for White students. In particular, columns (2) and (3) show no significant

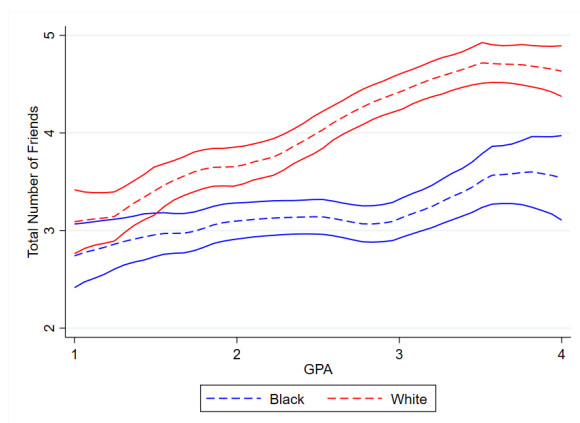
^{A.1} Specifically, in schools with Black student shares ranging from 0%-22%, 23%-40%, 41%-74%, and 75%-100%, the gaps in shares with GPA of 3-4 between Black and White students are 19%, 19%, 13%, and 1%, respectively.

relationship between $dGPA \times BB$ (the GPA difference interacted with a Black–Black pair indicator) and friendship formation in schools where White students are the majority. In contrast, we observe positive and significant effects in schools where Black students make up more than 40% of the population. Notably, the degree of racial homophily, measured by BB , varies with the proportion of Black students. As the share of Black students increases, the coefficient on BB tends to decline. These patterns remain consistent when we control for fixed effects of friendship senders and receivers, as shown in Appendix Table [A.6](#).

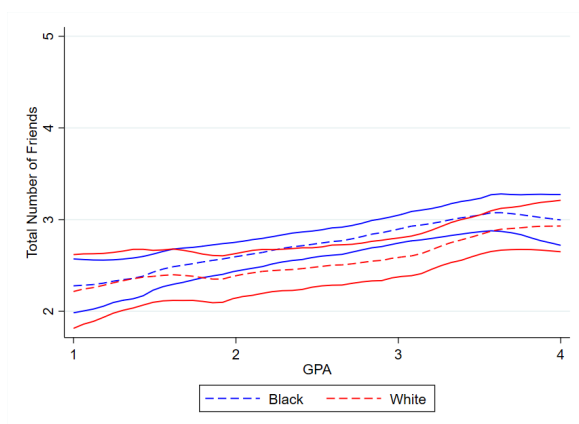
Fig. A.3. Number of Friends by GPA and Black Students' Share in Schools



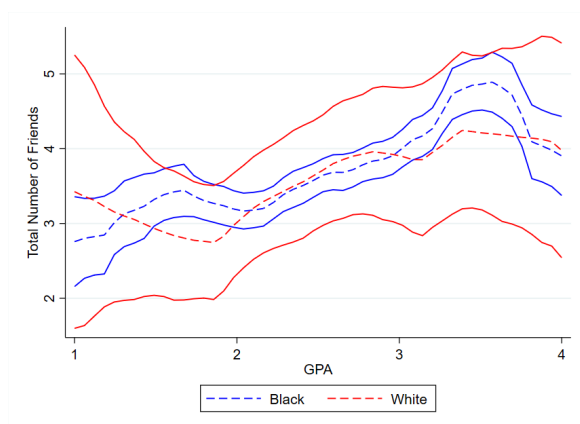
(a) Schools with 0%-22% Black enrollment



(b) Schools with 23%-40% Black enrollment



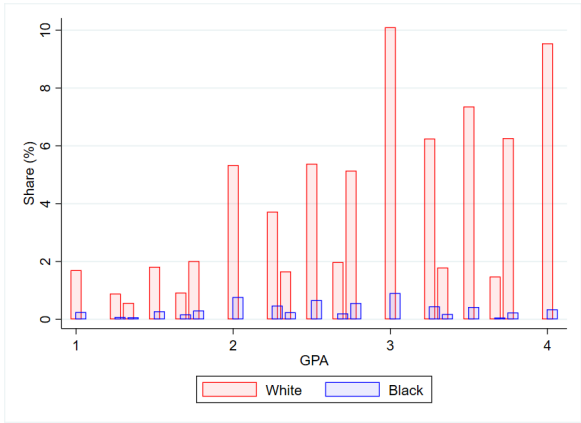
(c) Schools with 41%-74% Black enrollment



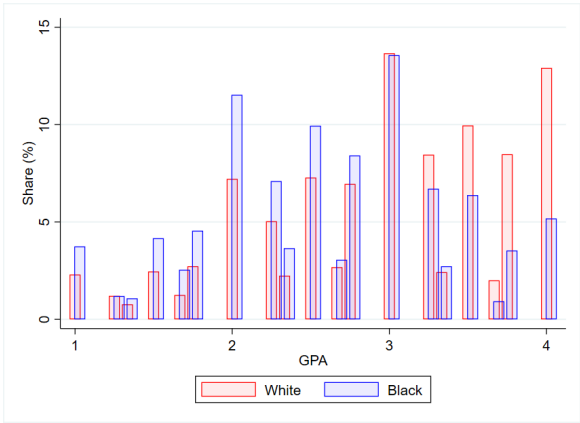
(d) Schools with 75%-100% Black enrollment

Notes. The figure depicts the locally smoothed means of the number of friends across GPA distribution, categorized by race and the share of Black students.

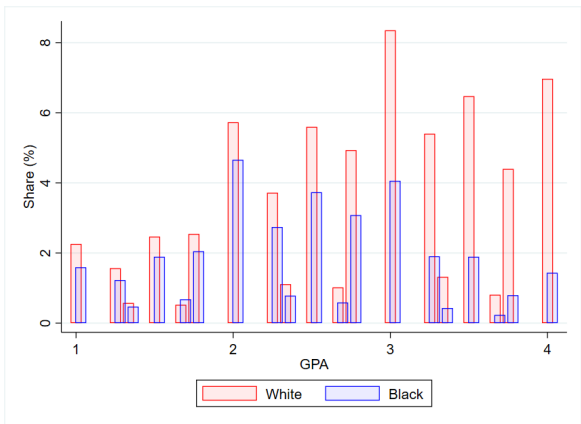
Fig. A.4. GPA Distribution by Race



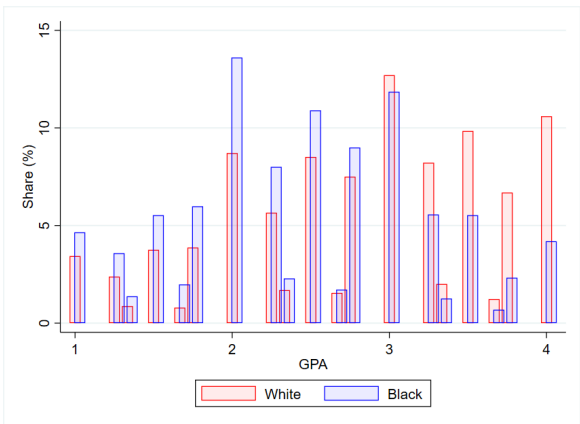
(a) Original Shares (Black 0%-22%)



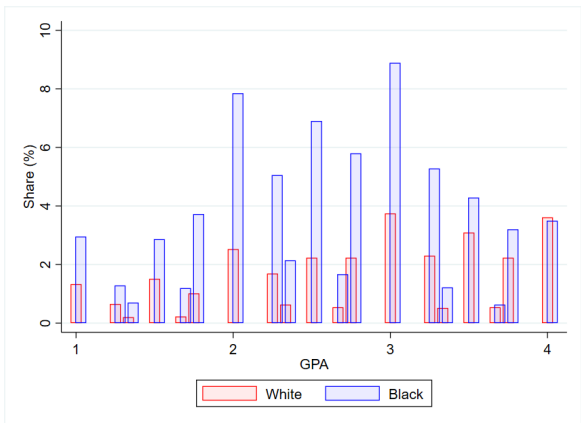
(b) Equal Shares (Black 0%-22%)



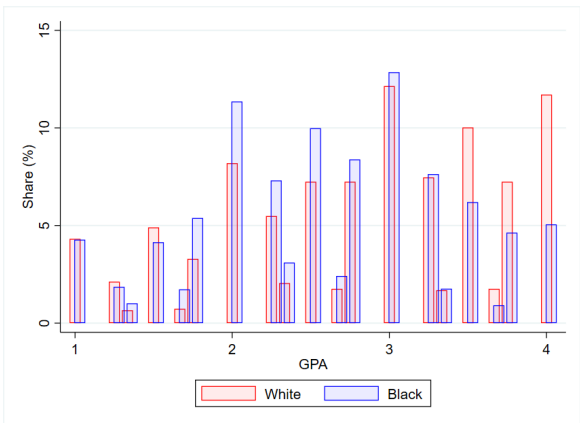
(c) Original Shares (Black 23%-40%)



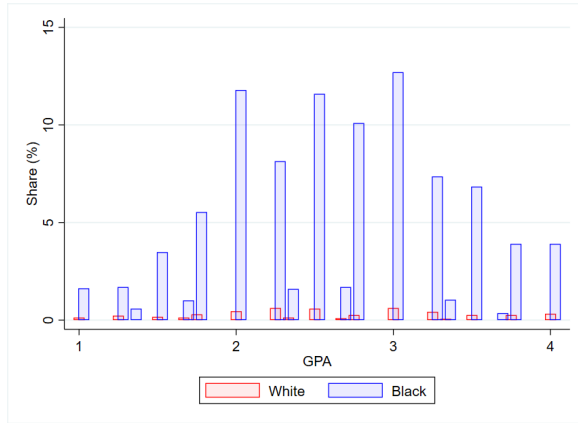
(d) Equal Shares (Black 23%-40%)



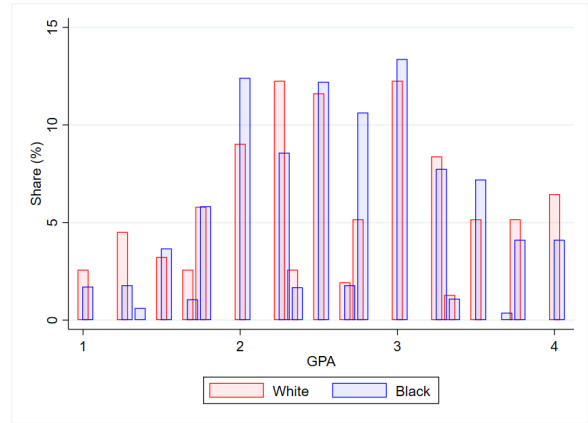
(e) Original Shares (Black 41%-74%)



(f) Equal Shares (Black 41%-74%)



(g) Original Shares (Black 75%-100%)



(h) Equal Shares (Black 75%-100%)

Notes. The figures illustrate the distributions of average GPA for Black and White students, separately by proportion of Black students in school. The red bar indicates the density of the average GPA of White students. The blue bar indicates the density of the average GPA of Black students. The sub-figures (a), (c), (e), and (g), illustrate the average GPA distribution with original Black and White shares. The sub-figures (b), (d), (f), and (h), illustrate the average GPA distribution after equalizing Black and White shares.

Table A.5: Friendship Formation by Race and Achievement: Across Black Student Share

	Dependent variable : Probability of i nominating j as a friend (%)				
	(1) All	(2) 0-22%	(3) 23-40%	(4) 41-74%	(5) 75-100%
$dGPA$	-0.189*** (0.022)	-0.179*** (0.022)	-0.284*** (0.062)	-0.267*** (0.032)	-1.361*** (0.116)
$dGPA \times BB$	0.017 (0.045)	-0.095 (0.075)	0.044 (0.032)	0.187*** (0.032)	1.152*** (0.127)
$dGPA \times BW$	0.158*** (0.022)	0.144*** (0.022)	0.267*** (0.061)	0.239*** (0.040)	1.317*** (0.106)
$dGPA \times WB$	0.158*** (0.023)	0.145*** (0.022)	0.265*** (0.063)	0.245*** (0.034)	1.322*** (0.098)
BB	0.250 (0.171)	1.212*** (0.285)	0.300** (0.128)	-0.513*** (0.122)	-5.207*** (0.186)
BW	-0.627*** (0.061)	-0.474*** (0.048)	-1.000*** (0.208)	-0.983*** (0.144)	-5.927*** (0.223)
WB	-0.603*** (0.060)	-0.448*** (0.047)	-0.987*** (0.213)	-0.952*** (0.128)	-5.838*** (0.259)
Constant	0.862*** (0.129)	0.741*** (0.098)	0.935*** (0.276)	1.701*** (0.244)	6.103*** (0.301)
Observations	36,183,256	26,910,900	4,993,870	2,743,852	1,534,634
Mean	0.513	0.509	0.539	0.400	0.686
Controls	X	X	X	X	X

Notes. The table presents coefficients and standard errors from estimation of equation (1), separately by shares of Black students. Units of observation are directed dyads, based on non-Hispanic Black and non-Hispanic White students who reported GPAs and individual characteristics (age, gender, and race). The dependent variable takes the value of 1 if student i nominated student j as a friend. $dGPA$ indicates a difference in average GPA within pairs of students. B indicates Black students, and W indicates White students. “Mean” refers to the average probability of forming friendships. “Controls” indicate the controlling demographic variables including gender and age of senders and receivers, difference in age within pairs, and school fixed effects. The schools are divided into four quantile groups as follows: the first quantile (Q1) includes schools with Black student shares ranging from 0% to 22%; the second quantile (Q2), 23% to 40%; the third quantile (Q3), 41% to 74%; and the fourth quantile (Q4), 75% to 100%. Standard errors are clustered at the school level. *** $p < .01$, ** $p < .05$, * $p < .1$

Table A.6: Friendship Formation by Race and Achievement: Across Black Student Proportion with Individual Fixed-Effects

	Dependent variable : Probability of i nominating j as a friend (%)				
	(1) All	(2) 0-22%	(3) 23-40%	(4) 41-74%	(5) 75-100%
$dGPA$	-0.259*** (0.028)	-0.248*** (0.029)	-0.359*** (0.078)	-0.269*** (0.035)	-1.406*** (0.124)
$dGPA \times BB$	0.038 (0.049)	-0.080 (0.075)	0.094** (0.039)	0.146*** (0.031)	1.136*** (0.120)
$dGPA \times BW$	0.200*** (0.025)	0.165*** (0.025)	0.295*** (0.067)	0.221*** (0.039)	1.298*** (0.106)
$dGPA \times WB$	0.193*** (0.026)	0.154*** (0.024)	0.281*** (0.067)	0.229*** (0.037)	1.354*** (0.097)
Constant	0.938*** (0.051)	1.154*** (0.070)	0.870*** (0.056)	0.097 (0.111)	-4.600*** (0.244)
Observations	36,183,256	26,910,900	4,993,870	2,743,852	1,534,634
Mean	0.513	0.509	0.539	0.400	0.681

Notes. The table presents coefficients and standard errors from estimation of equation (1), separately by shares of Black students. Units of observation are directed dyads, based on non-Hispanic Black and non-Hispanic White students who reported GPAs and individual characteristics (age, gender, and race). The dependent variable takes on the value of 1 if student i nominated student j as a friend. $dGPA$ indicates a difference in average GPA within the pairs of students. B indicates Black students, and W indicates White students. “Mean” refers to the average probability of forming friendships. The schools are divided into four quantile groups as follows: the first quantile (Q1) includes schools with Black student shares ranging from 0% to 22%; the second quantile (Q2), 23% to 40%; the third quantile (Q3), 41% to 74%; and the fourth quantile (Q4), 75% to 100%. Standard errors are clustered at the school level. *** $p < .01$, ** $p < .05$, * $p < .1$

A.5 Additional Simulation Outcomes

We provide additional details and results for the simulation analysis. We define the number of friends of student j as follows:

$$n_j = \sum_{i \in s} (G_{ij}) \quad (\text{A.1})$$

where s is the school that students i and j are in. Based on the estimated parameters in equation (1), the number of Black and White students, and the GPA distributions, we compute the number of simulated friends ($\tilde{n}$) for a student with GPA of x with the following equation:

$$\begin{aligned} \tilde{n} &= n(x, \hat{\alpha}, \hat{\beta}; N_B, N_W, f_B, f_W) \\ &= \sum_{s \in E} N_B^s \times \int_{lb}^{ub} \left[(\hat{\alpha}_0^s + \hat{\alpha}_{BB}^s B + \hat{\alpha}_{WB}^s W) + (\hat{\beta}_0^s + \hat{\beta}_{BB}^s B + \hat{\beta}_{WB}^s W) |x - z| \right] f_B^s dz \\ &\quad + \sum_{s \in E} N_W^s \times \int_{lb}^{ub} \left[(\hat{\alpha}_0^s + \hat{\alpha}_{BW}^s B) + (\hat{\beta}_0^s + \hat{\beta}_{BW}^s B) |x - z| \right] f_W^s dz, \end{aligned} \quad (\text{A.2})$$

where for each school s , N_B^s and N_W^s are the number of Black and White students, respectively. B and W are indicator variables taking the value 1 if a student is Black and White, respectively. $f_B^s(z)$ and $f_W^s(z)$ are the GPA distribution for Black and White students in school s . $\hat{\alpha}^s$ and $\hat{\beta}^s$ are the estimated parameters from equation (1) that capture the baseline probability of forming friendships and the interaction between GPA and race for the respective race pair. E represents the set of schools that contains all coefficients $\hat{\alpha}^s$ and $\hat{\beta}^s$. For low-achieving friends, lb and ub are designated as 1 and 3, respectively, while for high-achieving friends, lb and ub are set to 3 and 4.

The term $(\hat{\alpha}_0^s + \hat{\alpha}_{BB}^s B + \hat{\alpha}_{WB}^s W) + (\hat{\beta}_0^s + \hat{\beta}_{BB}^s B + \hat{\beta}_{WB}^s W) |x - z|$ represents the likelihood of forming friendships with Black peers when GPA distance is $|x - z|$. The term $(\hat{\alpha}_0^s + \hat{\alpha}_{BW}^s B) + (\hat{\beta}_0^s + \hat{\beta}_{BW}^s B) |x - z|$ represents the same likelihood for friendships with White peers. We calculate the $\tilde{n}$ for all GPAs x and compare them with the number of friends in the data.

We present the number of friends from the data in Figure A.5 (a) and from simulation in Figure A.5 (b). In Figure A.5 (a), we present the number of friends by race (n_j^W and n_j^B) and by friends' achievement, where the achievement is divided into high and low types. As GPA increases, both Black and White students are more likely to receive friendship nominations from

high-GPA students, but the slope is steeper for White students, creating a larger gap for students with high GPAs. Conversely, they are less likely to receive nominations from low-GPA students as GPA increases, but the change in the difference between Black and White students is not as significant as the change in nominations from high-GPA students.

Figure A.5 (b) displays the number of simulated friends for Black and White students ($\tilde{n}^B$ and $\tilde{n}^W$) separately by friends' achievement. The distribution of the simulated number of friends captures both qualitatively and quantitatively the patterns of the distribution in the data. We also calculate the average number of friends by GPA ranges 1-2, 2-3, and 3-4 for both data and simulation in Appendix Table A.7.^{A.2} Figure A.5 (c) presents a comparison between data and simulation regarding the racial difference in the number of high- and low-achieving friends (i.e., $n_j^W - n_j^B$ and $\tilde{n}^W - \tilde{n}^B$). Regardless of students' GPA distribution and friends' achievement, the racial gap is similar for the data and simulation, confirming the model's goodness of fit. As detailed in Appendix A.5, the difference in the number of friends between our simulation and the data is consistently less than 0.2 across all GPA ranges.

We calculate the racial friendship gap relative to the lowest GPA as follows:

$$\begin{aligned} d(x, \hat{\alpha}, \hat{\beta}; N_B, N_W, f_B, f_W) = & \tilde{n}^W(x, \hat{\alpha}, \hat{\beta}; N_B, N_W, f_B, f_W) - \tilde{n}^B(x, \hat{\alpha}, \hat{\beta}; N_B, N_W, f_B, f_W) \\ & - [\tilde{n}^W(x=1, \hat{\alpha}, \hat{\beta}; N_B, N_W, f_B, f_W) - \tilde{n}^B(x=1, \hat{\alpha}, \hat{\beta}; N_B, N_W, f_B, f_W)]. \end{aligned} \quad (\text{A.3})$$

In Figure 3, we compare the simulated number of high- and low-achieving friends before and after distributional changes. The solid blue line represents the difference in the simulated number of friends between White and Black students across the GPA distribution (i.e., $d(\hat{\alpha}, \hat{\beta}; N_B, N_W, f_B, f_W)$). The red dashed line depicts the difference in the simulated number of friends after Black and White student shares are equalized (i.e., $d(\hat{\alpha}, \hat{\beta}; N^*, N^*, f_B, f_W)$), where $N^* = (N_B + N_W)/2$, and the black short dashed line represents the gap after substituting the White GPA distribution for the Black GPA distribution, as well as the equal shares (i.e., $d(\hat{\alpha}, \hat{\beta}; N^*, N^*, f_W, f_W)$).

^{A.2} An observed dissimilarity between the data and simulation is that high-achieving students tend to have fewer friends in the simulation for both Black and White students. This discrepancy might indicate an unaccounted premium for high achievers, which our simple model does not consider. However, given our focus on racial comparisons, the omission of this factor, which affects both racial groups similarly, is unlikely to pose a significant issue in our racial comparison.

Appendix Table A.7 displays the number of friends categorized by students' GPA (1-2, 2-3, 3-4), students' race (Black, White), and friends' GPA (high-achieving, low-achieving). Furthermore, we illustrate the disparity in the number of friends between Black and White students, as well as the difference between data and simulation regarding the racial gap in friendship counts.

Appendix Table A.8 provides numerical values for the difference in the number of friends in each counterfactual analysis, complementing Figure 3.

Appendix Figure A.6 displays the total number of friends gap, including both high- and low-achieving friends.

Appendix Figure A.7 illustrates the friendship gap in the original simulation and each counterfactual scenario without normalization, where the gap is set to zero at the lowest GPA. In counterfactual 1, we equalize the proportions of Black and White students, resulting in the near elimination of the racial gap in high-achieving friends, while the gap in low-achieving friends becomes around negative 1.0. This suggests that Black students are expected to have the same number of high-achieving friends and approximately one more low-achieving friend compared to White students. In counterfactual 2, we further equalize their GPA distributions, presenting that Black students have approximately 0.5 more high- and low-achieving friends.

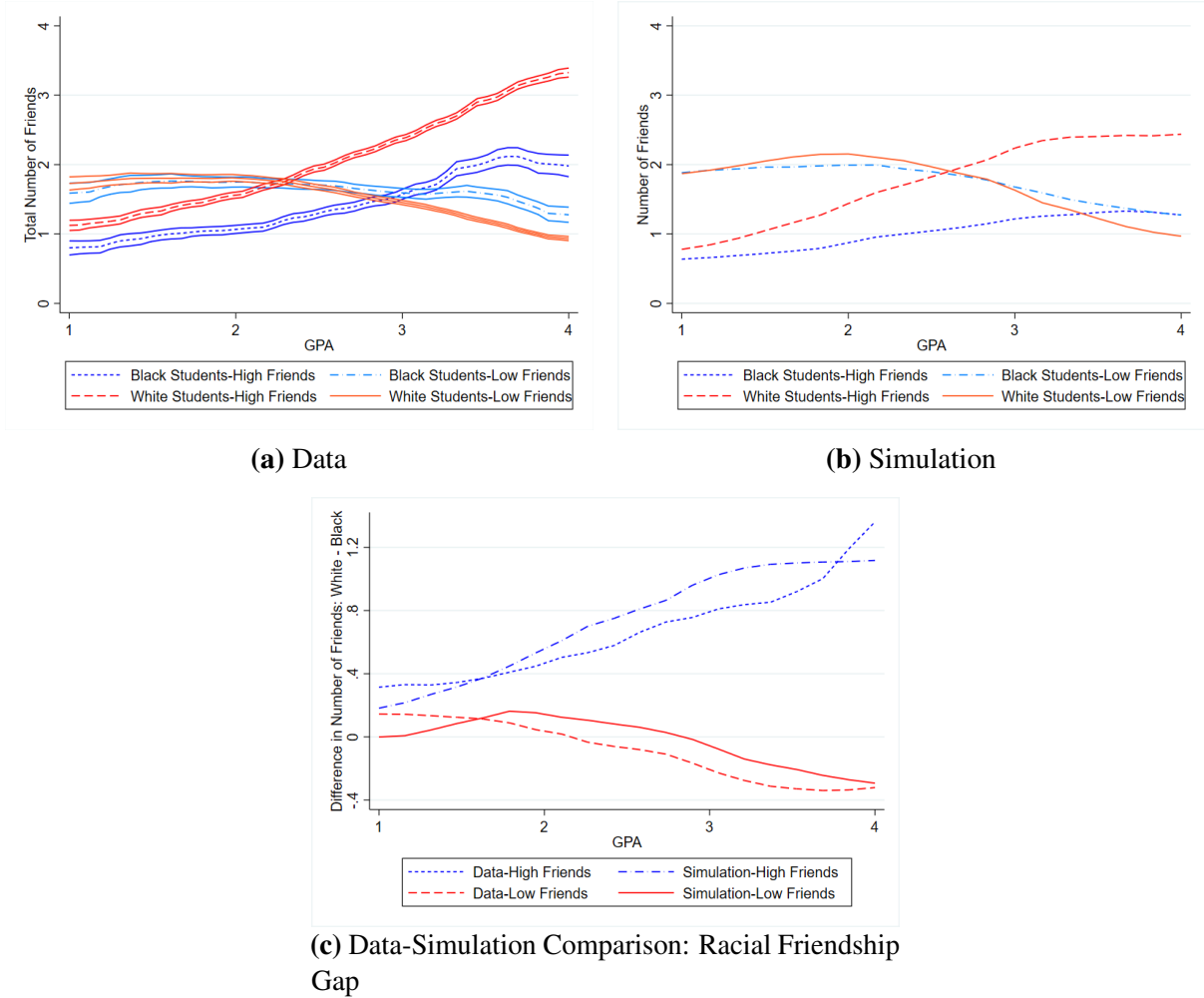
However, it is important to note that the more friends of Black students compared to White in the counterfactuals may be influenced by the higher level of racial homophily due to the smaller proportion of Black students. In our regression model (equation (1)), the overall level of racial homophily is determined by the probability of forming friendships among all possible pairs. Since the number of friends is constrained to a certain level (i.e., congestion), the overall level of homophily may be overestimated when the number of Black students is limited.^{A.3} Therefore, even if our estimation suggests that Black students have more friends, we focus on relative levels in the baseline analysis where overall homophily cannot exert influence. If, in reality, there are racial differences in the underlying preferences for friendship formation—and Black students tend to form more friendships—then the reduction in the friendship gap observed in

^{A.3} For example, having 10 Black friends among 20 Black peers results in a probability of 0.5. If there are 40 Black students, a student would need to have 20 Black friends to achieve the same probability. However, it is unrealistic to expect individuals to have such a large number of friends due to time and resource constraints.

our counterfactuals may be underestimated.

In Appendix Figure [A.9](#), we present results of counterfactual analyses using representative coefficients derived from all schools and schools with Black enrollment percentages between 23-40%, as presented in Table [1](#) or [A.5](#). In all instances, we observe a reduction in the racial gap in the number of friends when the Black-White proportion and GPA distributions are equalized. Notably, the explained gap for high-achieving students is larger than the baseline results in all cases. This mitigates concerns regarding the potential impact of student distributions on coefficients. Overall, we take a consistently conservative approach in estimating the reduction in the friendship gap in our counterfactual analyses.

Fig. A.5. Number of Friends from Data and Simulation



Notes. Figure (a) shows the number of friends across the GPA distribution, separately by race (Black and White) of the friendship receiver and achievement (high and low) of the sender. High-achieving students are defined as those with a GPA of at least 3. The short dashed blue line and dashed light blue line represent the number of high- and low-achieving friends of Black students, respectively. The dashed red line and solid orange line represent the number of high- and low-achieving friends of White students, respectively. Figure (b) shows the simulated number of friends for Black and White students, differentiated by the sender's achievement following equation (2). The line styles in Figure (b) correspond to those in Figure (a). Figure (c) compares the Black and White friendship gap observed in the data (Figure (a)) with the corresponding simulation results (Figure (b)). This comparison is done separately for high- and low-achieving friends. The short dashed blue line and dashed-dotted blue line indicate the racial gap in the number of high-achieving friends from the data and simulation, respectively. The dashed red line and the solid red line indicate the racial gap in the number of low-achieving friends from the data and simulation, respectively.

Table A.7: Number of friends by Type of Friends and GPA

	Black Students			White Students		Difference (White - Black)	
	(1) Low-Perform Friends	(2) High-Perform Friends	(3) Low-Perform Friends	(4) High-Perform Friends	(5) Low-Perform Friends	(6) High-Perform Friends	
Data							
GPA 1-2	1.71	0.98	1.79	1.37	0.08	0.39	
GPA 2-3	1.66	1.37	1.60	2.10	-0.06	0.73	
GPA 3-4	1.51	1.98	1.10	3.07	-0.41	1.09	
Simulation Results							
GPA 1-2	1.98	0.77	2.08	1.18	0.10	0.41	
GPA 2-3	1.87	1.11	1.87	2.00	0.00	0.89	
GPA 3-4	1.44	1.28	1.15	2.44	-0.29	1.16	
Difference (Data - Simulation)							
GPA 1-2	-0.27	0.21	-0.29	0.19	-0.02	-0.02	
GPA 2-3	-0.21	0.26	-0.27	0.10	-0.06	-0.16	
GPA 3-4	0.07	0.70	-0.05	0.63	-0.12	-0.07	

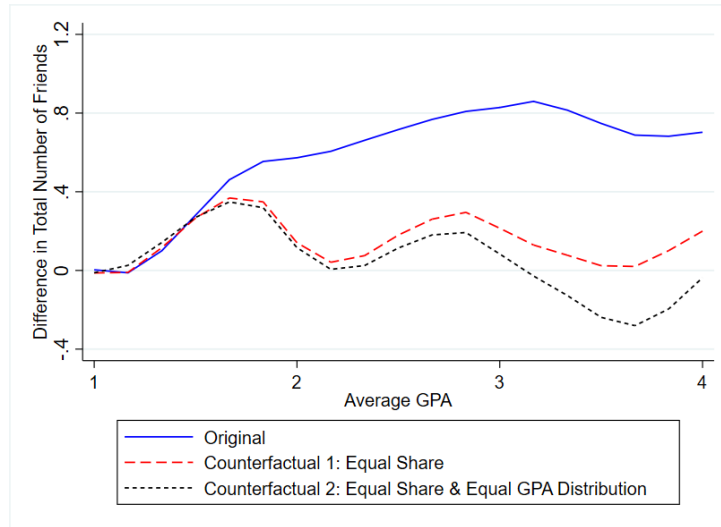
Notes. This table reports the number of friends formed, broken down by the race of friendship receivers and the achievement levels of both receivers and senders. Receiver achievement is categorized into GPA ranges of 1-2, 2-3, and 3-4, while sender achievement is grouped into low (1-3) and high (3-4). The Data panel presents the average number of friends observed in the dataset, and the Simulation Results panel shows the corresponding simulated values.

Table A.8: Racial Difference (White - Black) in the Number of Simulated Friends Before and After Composition Changes

Panel A: High-Achieving Friends			
	GPA 1-2	GPA 2-3	GPA 3-4
Simulation	0.31	0.75	1.05
Counterfactual 1: Equal Share	0.17	0.21	0.16
Counterfactual 2: Equal Share & Equal GPA Dist.	0.16	0.11	0.07
Panel B: Low-Achieving Friends			
	GPA 1-2	GPA 2-3	GPA 3-4
Simulation	0.08	0.01	-0.26
Counterfactual 1: Equal Share	0.01	-0.01	-0.01
Counterfactual 2: Equal Share & Equal GPA Dist.	0.03	-0.01	-0.15

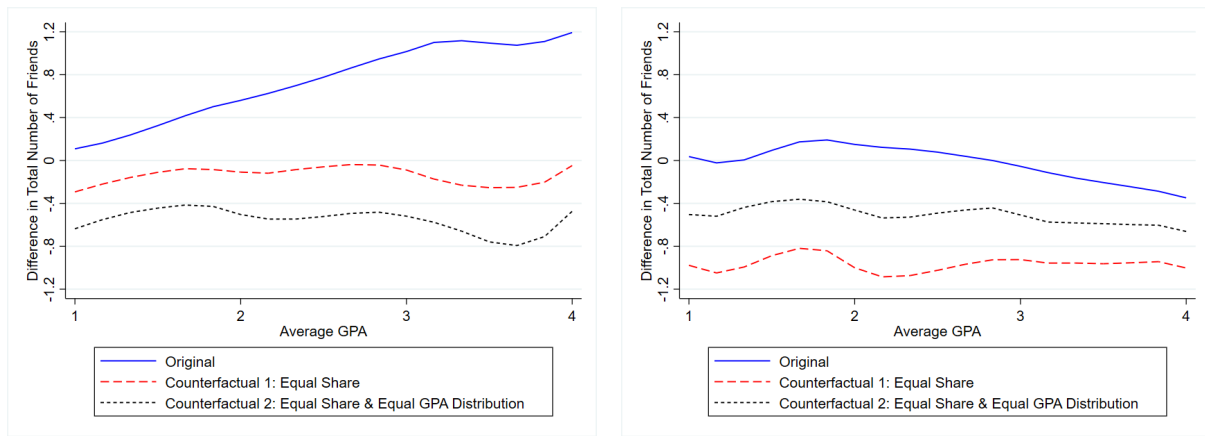
Notes. The table illustrates the difference in the number of simulated friends between Black and White students, with standardization of 0 at the lowest GPA level. The simulated numbers are computed using equation (2). Panel (a) and (b) present the difference in the number of high-achieving and low-achieving friends separately. In the table, the simulation presents the original difference in the simulated number of friends. Counterfactual 1 presents the difference assuming an equal proportion of Black and White students. Counterfactual 2 presents the difference assuming both an equal share of Black and White students and an identical GPA distribution for Black and White students.

Fig. A.6. Racial Difference (White - Black) in the Number of Simulated Friends by GPA



Notes. The figure illustrates the difference in the number of simulated friends between Black and White students, calculated as White minus Black and standardized to 0 at the lowest GPA level following equation (2). The solid blue line depicts the original difference in the simulated number of friends. The dashed red line represents the difference assuming an equal proportion of Black and White students. The short-dashed black line represents the difference assuming both an equal share of Black and White students and an identical GPA distribution for Black and White students.

Fig. A.7. Racial Difference (White - Black) in the Number of Simulated Friends by GPA and Friend Achievement Level: Without Normalization

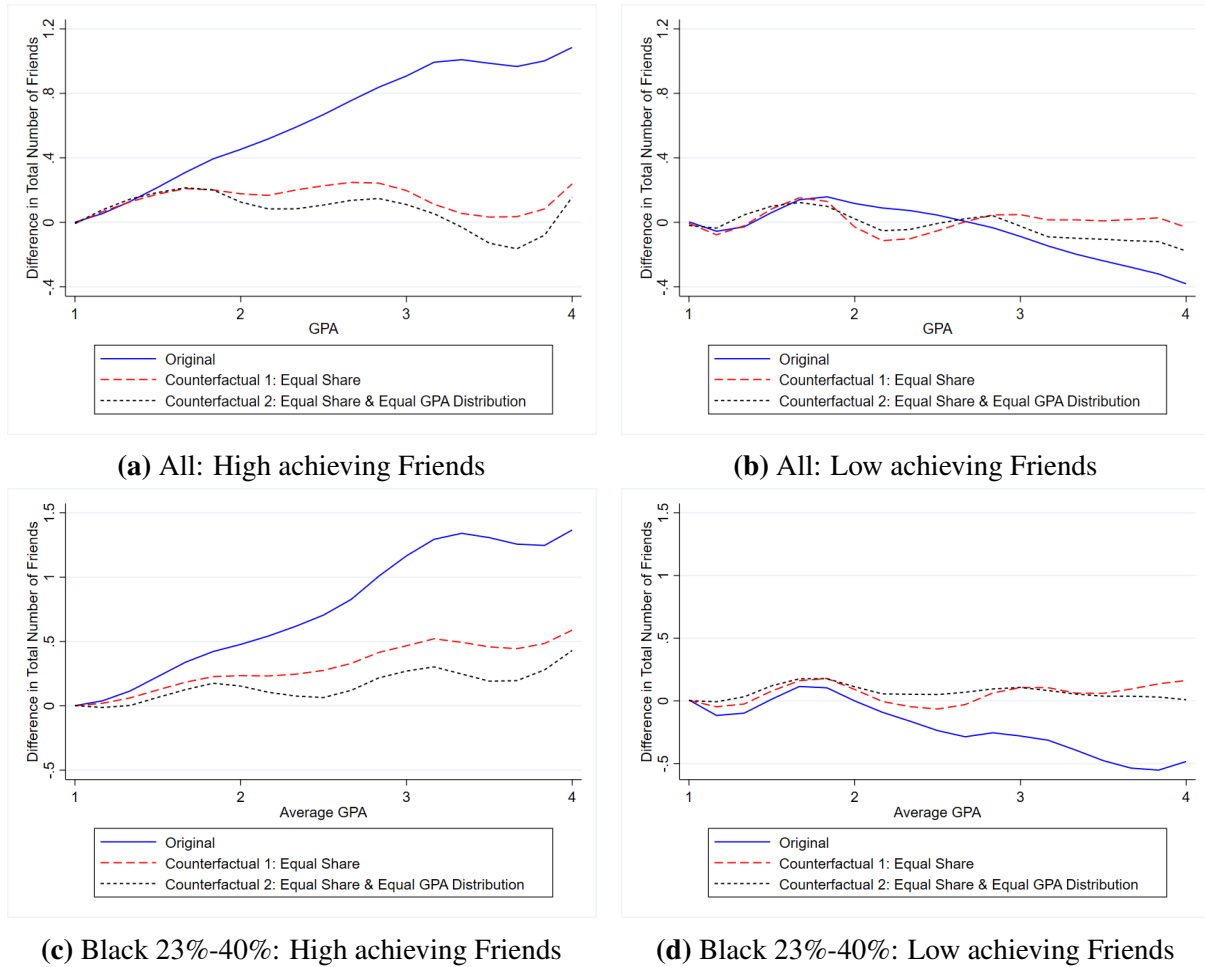


(a) High achieving Friends

(b) Low achieving Friends

Notes. The figure illustrates the difference in the number of simulated friends between Black and White students, calculated as White minus Black, following equation (3) without normalization. The simulated numbers are computed based on equation (2). Figures (a) and (b) present the difference in the number of high-achieving and low-achieving friends, separately. The solid blue line depicts the original difference in the simulated number of friends. The dashed red line represents the difference assuming an equal proportion of Black and White students. The short-dashed black line represents the difference assuming both an equal share of Black and White students and an identical GPA distribution for Black and White students.

Fig. A.9. Racial Difference (White - Black) in the Number of Simulated Friends by GPA and Friend Achievement Level with Coefficients from Representative Schools



Notes. The figure illustrates the difference in the number of simulated friends between Black and White students, calculated as White minus Black and standardized to 0 at the lowest GPA level following equation (2). Specifically, we use coefficients from the representative schools calculated in Table 1 or A.5. Figure (a) and (b) use coefficients from all schools, and Figure (c) and (d) use coefficients from schools with Black enrollment between 23-40%. In the graph, the solid blue line depicts the original difference in the simulated number of friends. The dashed red line represents the difference assuming an equal proportion of Black and White students. The short-dashed black line represents the difference assuming both an equal share of Black and White students and an identical GPA distribution for Black and White students.

A.6 Additional Earnings Outcomes

We further restrict our sample for labor market analysis to include individuals who report measures of cognitive skills, social skills, education level, and earnings. Consistent with Lleras-Muney et al. (2025), cognitive skills are assessed using the Add Health Picture Vocabulary Test (AHPVT) score administered in Wave 1, while social skills are gauged through self-reported extroversion levels collected in Wave 2.^{A.4} Earnings are measured as total earnings from wages or salary in the previous year. We utilize earnings data from Waves 4, corresponding to ages 24-34, on average 29 years old. Respondents who responded “do not know” to the earnings question were presented with twelve earnings categories, and we approximate their earnings by using the midpoint of the selected range. We drop individuals who reported zero earnings (7%). On average, individuals in our sample earned approximately \$37,300 in the preceding year. See Appendix Table A.9 for summary statistics of the earnings sample.

To estimate the causal return to the number of friends, Lleras-Muney et al. (2025) follow the identification strategy of Cicala, Fryer, and Spenkuch (2018) and Murphy and Weinhardt (2020), who leverage within-cohort variation in relative position—such as academic rank or age distribution—to study peer effects and behavioral outcomes. Following Lleras-Muney et al. (2025), we exploit variation in (race-specific) age distance—the average age gap between a student and their peers in the same school and grade—while controlling for individual age and cohort-mean age. This approach isolates residual variation in age distance that is not mechanically determined by a student’s own age but instead arises from differences in the age composition of cohorts. Our identifying assumption is that, conditional on a rich set of controls, age distance affects outcomes only through its influence on peer interactions and educational experiences—satisfying the exclusion restriction.

To empirically support this assumption, Lleras-Muney et al. (2025) conduct a series of placebo tests using a range of predetermined characteristics, including parental education, parental and own nativity, religion, birthweight, breastfeeding history, height, and disability

^{A.4} The survey question is “You are shy?”, and the choices are “strongly agree, agree, neither agree nor disagree, disagree, strongly disagree”. Individuals choosing last three categories are defined as extrovert. Due to the survey design, 26% of the individuals in the dataset lack this information. Consequently, we impute this measure and incorporate a dummy variable to indicate its absence.

status. These variables are largely determined early in life and are known predictors of long-run outcomes such as earnings. They find that age distance does not significantly predict any of these characteristics after adjusting for basic controls; the coefficients are consistently small and statistically insignificant across all specifications. Furthermore, a joint F-test regressing age distance on the full set of predetermined variables shows no meaningful relationship, indicating that age distance is unrelated to these early-life characteristics. These results provide empirical support for the validity of the exclusion restriction (see their Tables A.3 and A.4).

Appendix Table A.10 presents the first-stage results of our IV estimation.^{A.5} In column (1), we use age distance to peers in the same school and grade as our instrument. The distance is the average of the absolute difference between all possible pairs in the group with a student. In column (2), we disaggregate the age distance into three components: distance to Black, White, and other than Black and White peers in the same school and grade. Thus, each student has three age-distance variables. The rationale is that the age distance measure may predominantly reflect distance to White students, even if Black students are more inclined to form friendships with other Black students.

In column (2), we interact age distance with own-race indicators to examine potential racial differences in sensitivity to friendships. Students are more sensitive to age differences when they form friendships with their same-race peers. To assess instrument strength, we report joint weak identification diagnostics for the multiple endogenous variables, *NumFriend* and *Black* $\times$ *NumFriend*. The Kleibergen–Paap *rk* Wald F statistic is 6.43. For reference, the Stock–Yogo critical value for a 10% maximal IV relative bias is 7.77. Because the Kleibergen–Paap F is below this threshold, the instruments are potentially weak, and the 2SLS estimates should be interpreted with caution.

In column (3), we disaggregate the age distance into low- and high-achieving peers, and interact these with a low-achieving dummy to consider homophily in achievement. Again, we find stronger sensitivity to the same group. The Kleibergen–Paap *rk* Wald F statistic (19.03) exceeds the Stock–Yogo 5% critical value (11.04), indicating that weak instruments are unlikely to be a major concern. Columns (1), (2), and (3) are matched to columns (3) and (4) of Table

^{A.5} Our analysis sample is smaller than that of Lleras-Muney et al. (2025) because we use race-specific age distance measures, which exclude approximately 1,500 students.

??, and column (6) of Appendix Table A.11, respectively. In all cases, we observe a strong relationship between friendship and age distance variables.

Table A.11 presents OLS and 2SLS results. In columns (1)-(3), we present OLS estimates for the return to the number of friends. Since our focus is on the racial gap, we estimate returns separately for Black and White students. Column (2) further distinguishes between high- and low-achieving friends. We find no evidence of different returns to the number of friends between Black and White students, regardless of whether friends are high-achieving or low-achieving. Column (3) presents estimation results for high- and low-achieving students separately. The results indicate that low-achieving students have smaller returns from additional friends, which might reflect differences in the friends they have due to homophily. While having one more friend correlates with a 2.9% increase in earnings for high-achieving students, it corresponds to a 1.5% increase for low-achieving students.

In columns (4)-(6), IV regression estimates are presented using overall age distance and race-specific age distance as instruments, respectively. Overall age distance is calculated based on age distance among all possible pairs in a school-grade following Lleras-Muney et al. (2025). The race-specific distance is calculated separately for Black and White peers to account for racial homophily in friendship formation. In column (4), we estimate the overall return to one more friend for students of all races. In column (5), we estimate the return separately for Black and White students using race-specific distance as the IV. We find a larger return to the number of friends in both cases while results are not statistically significant at the 5% level because of large standard errors. Once again, we do not find that returns differ between Black and White students. In column (6), we also present results dividing students based on achievement. We find that high-achieving students gain a higher return from having one more friend, while the difference is not large. In their various specifications, Lleras-Muney et al. (2025) estimate the bounds of return to the number of friends as 6.50% to 13.67%. Since our estimates fall mostly within these bounds and are not precise, we use their bounds for our back-of-the-envelope calculation for the racial earnings gap. Lleras-Muney et al. (2025) use the same data source and define friendship in the same way as this paper, though their sample selection differs slightly. Their point estimate under a specification similar to Table ?? column (3) is 12.42%.

Our estimation results are consistent with those of Lleras-Muney et al. (2025), even when using race-specific age distance measures or dividing the sample by academic achievement. Their baseline estimate suggests a 12.4% return to friendship, with bounds ranging from 6.5% to 13.7%, assuming the return to education falls between 5% and 15% (see their Table 3). As in their study, our OLS estimates are downward biased, since the number of friends may be correlated with unobserved factors that also affect earnings. For example, behaviors like partying and drinking—often motivated by the desire to socialize (Feldman et al. 1999; Kuntsche et al. 2005)—may increase friendship ties but negatively impact long-term outcomes such as educational attainment or job performance.

Table A.9: Individual-Level Summary Statistics (Earnings Sample)

	White	Black	Other	All
GPA	2.9	2.6	2.9	2.8
Female (%)	51	58	51	53
Age	16	16	16	16
Number of Friends	3.80	2.97	2.88	3.30
High Performing ($\text{GPA} \geq 3$)	2.47	1.58	1.66	2.00
Low Performing ($\text{GPA} < 3$)	1.33	1.40	1.22	1.31
Cognitive Skills	106	97	98	101
Social Skills	0.47	0.44	0.38	0.43
Share (%)	44	21	35	100
Observations	3,957	1,949	3,175	9,081

Notes. The table presents individual-level summary statistics for White students, Black students, students of other races (including Hispanic, Asian, American Indian, and other races), and the full sample.

Table A.10: First-Stage: Effects of Age Distance on the Number of Friendships

<i>Outcomes are Number of Friends</i>	(1)	(2)	(3)
Age distance	-1.102*** (0.121)		
Age distance to Black Peers		0.030 (0.107)	
Age distance to White Peers		-0.305 (0.295)	
Age distance to Other Peers		-0.329 (0.251)	
Age distance to White Peers $\times$ White		-0.276 (0.184)	
Age distance to Black Peers $\times$ Black		-0.433** (0.209)	
Age distance to Other Peers $\times$ Other		-0.254** (0.103)	
Age distance to Low-Achieving Peers			-0.537 (0.376)
Age distance to High-Achieving Peers			-0.397 (0.388)
Age distance to Low-Achieving Peers $\times$ Low-Achieving			-0.317 (0.516)
Age distance to High-Achieving Peers $\times$ Low-Achieving			-0.074 (0.493)
F-stat of Homophily Measures	83.0	12.5	30.4
P-value	0.000	0.000	0.000
R^2	0.041	0.040	0.046
Observations	9,059	9,059	9,047
Controls	X	X	X

Notes. The table reports the first-stage results from the 2SLS estimation of equation 4. “Controls” refers to the inclusion of control variables such as age, sex, race, social skill measure, cognitive skill measure, and school and grade fixed effects. Age distance is the average age difference between a student and all other students in the same school and grade. Age distance to Black (White or Other) peers is the average age difference between a student and all Black (White or Other) students in the same school and grade. Black, White, and Other are dummy variables equal to one if a student is Black, White, or another race, respectively. Other includes Hispanic, Asian, American Indian, and all other races. Standard errors are clustered at the school level. *** $p < .01$, ** $p < .05$, * $p < .1$

Table A.11: Effects of the Number of Friends on Earnings: Full Table

	OLS			2SLS		
	(1)	(2)	(3)	(4)	(5)	(6)
Friends	0.024*** (0.004)		0.029*** (0.004)	0.146* (0.081)	0.123 (0.105)	0.117 (0.072)
Friends × Black	-0.002 (0.012)				0.034 (0.082)	
Friends × Other	0.002 (0.006)				0.022 (0.015)	
Friends × Low-Achieving			-0.014*** (0.004)			-0.019* (0.010)
High-Achieving Friends		0.029*** (0.004)				
Low-Achieving Friends		0.011* (0.006)				
High-Achieving Friends × Black		0.003 (0.013)				
Low-Achieving Friends × Black		-0.005 (0.014)				
High-Achieving Friends × Other		-0.004 (0.007)				
Low-Achieving Friends × Other		0.014 (0.009)				
Controls	X	X	X	X	X	X
IV				Overall	Racial	GPA
Observations	9,059	9,059	9,059	9,059	9,059	9,047

Notes. The table presents estimation results from equation (4). Columns (1)-(3) present OLS estimates, and columns (4)-(6) present 2SLS estimates in which the number of friends is endogenous. See A.6 for first-stage results. "Controls" stands for the inclusion of control variables, including age, sex, race, social skills, cognitive skills, and school and grade fixed effects. "Overall" indicates average age distance for all students in a school-grade. "Race-Specific" indicates the race-specific average age distance in school-grade. "GPA" indicates the achievement-specific, high- and low-achieving, age distance in school-grade. "Black" and "Other" indicate a dummy variable taking the value of 1 if the race is Black and other than Black and White, respectively. We use students of all races. Standard errors are clustered at the school level. *** $p < .01$, ** $p < .05$, * $p < .1$

B Sensitivity Analysis

Table B.1: Summary of Sensitivity Analysis

Robustness Check	Description	Key Finding
Alternative achievement measure	Use mother's education instead of GPA to address potential endogeneity from GPA being influenced by friendships (Appendix B.1)	Results are qualitatively similar
Within-club comparison	Restrict dyads to students participating in the same extracurricular activity to ensure comparable exposure (Appendix B.2)	Patterns of homophily and achievement sensitivity remain
Mutual friendships only	Use only reciprocated friendship nominations to test the robustness of the friendship measure (Appendix B.3)	Main findings hold
Asymmetric GPA effect	Distinguish between low-to-high and high-to-low GPA nomination directions to assess whether asymmetry matters (Appendix B.4)	Results remain consistent; asymmetry does not drive the main findings
Semiparametric GPA definition	Use binary indicators for high vs. low GPA instead of a continuous measure to explore differences across GPA groups (Appendix B.5)	Patterns of homophily persist
Inclusion of Hispanic students	Expand the sample to include Hispanic students to test generalizability of findings (Appendix B.6)	Core findings remain consistent

B.1 Using Socioeconomic Status Instead of GPA

In this subsection, we analyze the distribution of socioeconomic status (SES) groups and investigate friendship formation based on race and SES groups. We define socioeconomic status using two indicators of family background: the mother's educational attainment and whether the student lives with both parents. The high SES group includes students whose mothers have attained some college education or higher and who live with both parents. The remaining students are classified into the low SES group. Our definition of the high SES group aligns with the definition of the most advantaged group in Lundberg (2013).^{B.1}

By utilizing SES groups based on family backgrounds, we address concerns about the reverse causality between GPA and friendship formations. Instead of achievement affecting friendship formation, friendship formation may affect academic achievements. However, the measurement of the SES group is not affected by this issue as the SES group is a predetermined characteristic of students.

Appendix Figure B.1 illustrates the distribution of socioeconomic status (SES) for Black and White students. In Figure (a), White students comprise the majority in both SES categories while Black students represent the minority. In Figure (b), the distribution of SES for Black and White students is displayed after equalizing the shares of Black and White students. The figure presents that 30 percent of Black students are in the high SES category, whereas 40 percent of White students fall into the high SES category. Consequently, within their respective racial groups, Black students have a smaller pool of potential high SES friends compared to White students.

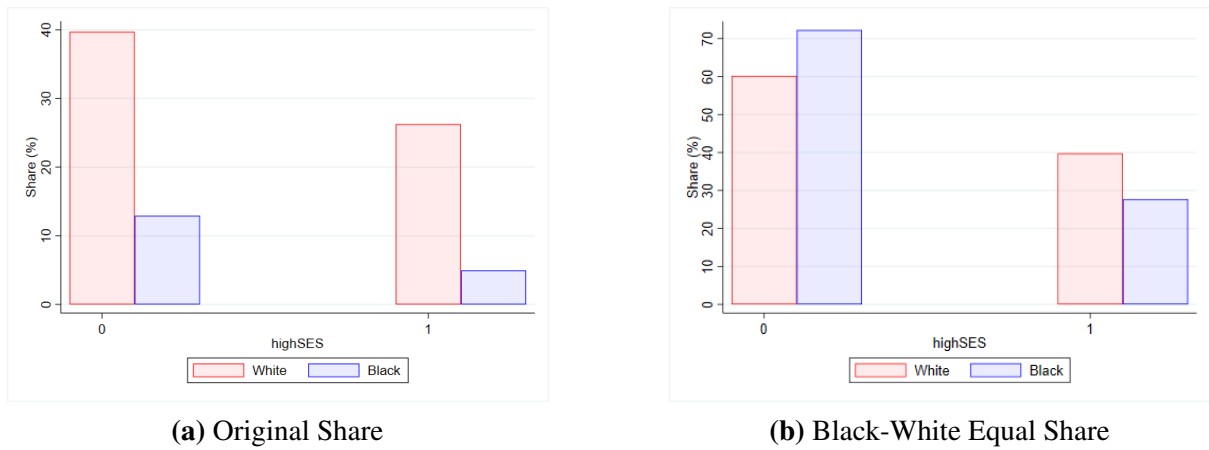
Appendix Table B.2 presents estimation results in equation (1) with and without covariates, using the SES group instead of GPA. In column (1), if the SES groups differ, the probability of friendship formation between two students decreases by 0.078 percentage points when the overall mean is 0.53 percent. In column (2), when we control for additional characteristics, including dyad-level characteristics (age and gender pairs) and demographic variables related to the senders' or receivers' genders and ages, the coefficient becomes more pronounced. In

^{B.1} Although Lundberg (2013) defines the most advantaged group as students with both biological parents and high maternal education, we use both parents instead as we cannot identify whether both parents of students are biological parents from the in-school data that we use for the analysis.

column (2), the results indicate that when the SES group differs, the probability of friendship formation between two students in the school decreases by 0.089 percentage points.

In column (3), we find that different SES matters similarly between pairs of White students and pairs of Black students. In column (4), we find similar results to those in column (3) after controlling for dyad-level and individual-level characteristics. In column (5), we control for sender and receiver fixed effects. Regardless of whether we control for individual fixed effects, we find that the difference in SES matters similarly in friendship formation for Black and White students. Moreover, the significance of the SES difference is more pronounced for within-race friendships compared to across-race friendships, regardless of the control variables. Overall, our findings remain consistent when using students' pre-determined socioeconomic status instead of GPA, helping to alleviate concerns about reverse causality.

Fig. B.1. SES Distribution by Race



Notes. The figure illustrates the distribution of SES for Black and White students. The sub-figure (a) illustrates the distribution of SES with original Black and White shares. The sub-figure (b) illustrates the distribution of SES after equalizing Black and White shares. High SES includes students in high maternal education and residence with both parents.

Table B.2: Friendship Formation by Race and SES

	Dependent variable : <i>i</i> nominating <i>j</i> as a friend (%)				
	(1)	(2)	(3)	(4)	(5)
DifferentSES	-0.078*** (0.014)	-0.089*** (0.013)	-0.083*** (0.019)	-0.090*** (0.013)	-0.110*** (0.015)
DifferentSES $\times$ <i>BB</i>			-0.022 (0.042)	-0.038** (0.017)	-0.005 (0.032)
DifferentSES $\times$ <i>BW</i>			0.074*** (0.020)	0.072*** (0.020)	0.051** (0.025)
DifferentSES $\times$ <i>WB</i>			0.063*** (0.021)	0.073*** (0.020)	0.061*** (0.023)
<i>BB</i>			0.211 (0.129)	0.267* (0.145)	
<i>BW</i>			-0.516*** (0.063)	-0.546*** (0.053)	
<i>WB</i>			-0.495*** (0.063)	-0.525*** (0.053)	
Constant	0.564*** (0.047)	0.705*** (0.103)	0.626*** (0.064)	0.839*** (0.129)	0.878*** (0.043)
Observations	30,156,736	30,156,736	30,156,736	30,156,736	30,156,736
Mean	0.530	0.530	0.530	0.530	0.530
Controls		X		X	X
Individual FE					X

Notes. The table presents coefficients and standard errors from estimation of equation (1). Units of observation are directed dyads, based on the non-Hispanic Black and non-Hispanic White students who reported mothers' education, residency with both parents, and individual characteristics (age, gender, and race). The dependent variable takes the value of 1 if student *i* nominates student *j* as a friend. "DifferentSES" is a binary variable equal to one if both students are in a different SES group. *B* indicates Black student and *W* indicates White student. *BW* is an indicator that the sender is a Black student and the receiver is a White student. "Mean" refers to the average probability of forming friendships. "Controls" indicate the controlling demographic variables including gender and age of senders and receivers, difference in age within pairs, and school fixed effects. "Individual FE" indicates that individual fixed effects of senders and receivers are controlled for. Standard errors are clustered at the school level. *** $p < .01$, ** $p < .05$, * $p < .1$

B.2 Friendship Formation by Race and Achievement: Within the Same School Club

In Appendix Table B.3, we examine whether the difference in GPA has varying implications for friendship formation when students are members of the same social club or not. The variable *Club* indicates whether the two students are both members of at least one social club. Our findings show that students who share membership in a club are more likely to form friendships compared to those who do not share club membership. Furthermore, we observe that the difference in GPA matters more on friendship formation for student pairs who are part of the same club ($Club \times dGPA$). In other words, when students belong to the same club—where they are expected to know one another—the likelihood of forming friendships decreases more as GPA differences increase. These findings address concerns that the formation of friendships is solely influenced by exposure rather than the active choice of students. In addition, we do not find a significant difference between White-White pairs and Black-Black pairs, regardless of their involvement in the social club ($Club \times dGPA \times BB$ and $dGPA \times BB$). Moreover, the significance of the achievement difference is more pronounced for within-race friendships compared to across-race friendships, regardless of the control variables.

Table B.3: Friendship Formation by Race and Achievement: Within School Clubs

	Dependent variable : <i>i</i> nominating <i>j</i> as a friend (%)	
	(1)	(2)
<i>Club</i> × <i>dGPA</i>	-0.355*** (0.046)	-0.353*** (0.041)
<i>Club</i> × <i>dGPA</i> × <i>BB</i>	0.103 (0.095)	0.116 (0.085)
<i>Club</i> × <i>dGPA</i> × <i>BW</i>	0.283*** (0.046)	0.288*** (0.044)
<i>Club</i> × <i>dGPA</i> × <i>WB</i>	0.285*** (0.045)	0.290*** (0.042)
<i>Club</i> × <i>BB</i>	-0.123 (0.230)	-0.106 (0.199)
<i>Club</i> × <i>BW</i>	-0.962*** (0.109)	-0.936*** (0.107)
<i>Club</i> × <i>WB</i>	-0.923*** (0.107)	-0.896*** (0.104)
<i>Club</i>	1.242*** (0.110)	1.096*** (0.098)
<i>dGPA</i>	-0.130*** (0.016)	-0.127*** (0.015)
<i>dGPA</i> × <i>BB</i>	0.012 (0.030)	0.002 (0.032)
<i>dGPA</i> × <i>BW</i>	0.113*** (0.016)	0.100*** (0.016)
<i>dGPA</i> × <i>WB</i>	0.108*** (0.016)	0.100*** (0.016)
<i>BB</i>	0.207 (0.125)	0.228 (0.140)
<i>BW</i>	-0.434*** (0.057)	-0.489*** (0.052)
<i>WB</i>	-0.413*** (0.057)	-0.470*** (0.052)
Constant	0.526*** (0.058)	0.592*** (0.125)
Observations	36,183,256	36,183,256
Mean	0.5	0.5
Controls		X

Notes. The table presents coefficients and standard errors from estimation of equation (1). Units of observation are directed dyads, based on the non-Hispanic Black and non-Hispanic White students who reported GPAs and individual characteristics (age, gender, and race). The dependent variable takes the value of 1 if student *i* nominates student *j* as a friend. *dGPA* indicates a difference in average GPA within a pair of students. *Club* is a binary variable equal to one if both students were in at least one extracurricular activity. *B* indicates Black students, and *W* indicates White students. *BW* is an indicator that the sender is a Black student and the receiver is a White student. “Mean” refers to the average probability of forming friendships. “Controls” indicate the controlling demographic variables including gender and age of senders and receivers, difference in age within pairs, and school fixed effects. Standard errors are clustered at the school level. *** $p < .01$, ** $p < .05$, * $p < .1$

B.3 Mutual Friendship Measures

In our dyadic analysis, we measure friendships based on directed numbers, where one student (i) directs another (j) as their friend. However, in Appendix Table B.4, we examine the sensitivity of our friendship measure by using a stricter definition of friendship: mutual friendship, where both students mutually direct each other as friends. While this stricter definition results in a smaller probability of forming friendships compared to the baseline measure, our findings are qualitatively unchanged. The difference in GPA matters again similarly between pairs of Black students and pairs of White students ($dGPA \times BB$). Additionally, we find that the difference in GPA matters less on across-race friendship formations compared to within-race friendships ($dGPA \times BW$ and $dGPA \times WB$).

Moreover, we alternatively define the mutual friendship measure, where if at least one student directs the other as a friend, both the $i - j$ and $j - i$ pairs are considered as forming friendships. The results are presented in Appendix Table B.5. This looser definition of friendship results in a higher probability of forming friendships compared to our baseline measure. However, our findings are qualitatively unchanged. The difference in GPA matters once again similarly between pairs of Black students and pairs of White students. We find that the difference in GPA matters less on across-race friendship formations compared to within-race friendships.

Table B.4: Friendship Formation by Race and Achievement: Mutual Friendship (Both)

Dependent variable : <i>i</i> nominating <i>j</i> as a friend (%)					
	(1)	(2)	(3)	(4)	(5)
<i>dGPA</i>	-0.095*** (0.010)	-0.078*** (0.008)	-0.112*** (0.013)	-0.093*** (0.011)	-0.131*** (0.015)
<i>dGPA</i> × <i>BB</i>			0.036* (0.021)	0.022 (0.022)	0.034 (0.025)
<i>dGPA</i> × <i>BW</i>			0.100*** (0.013)	0.086*** (0.012)	0.104*** (0.013)
<i>dGPA</i> × <i>WB</i>			0.100*** (0.013)	0.086*** (0.012)	0.104*** (0.013)
<i>BB</i>			0.002 (0.065)	0.026 (0.069)	0.928*** (0.108)
<i>BW</i>			-0.295*** (0.035)	-0.296*** (0.029)	0.000 (0.000)
<i>WB</i>			-0.295*** (0.035)	-0.296*** (0.029)	0.000 (0.000)
Constant	0.289*** (0.026)	0.088 (0.054)	0.335*** (0.036)	0.178*** (0.064)	0.373*** (0.021)
Observations	36,183,256	36,183,256	36,183,256	36,183,256	36,183,256
Mean	0.208	0.208	0.208	0.208	0.208
Controls		X		X	X
Individual FE					X

Notes. The table presents coefficients and standard errors from estimation of equation (1). Units of observation are directed dyads, based on the non-Hispanic Black and non-Hispanic White students who reported GPAs and individual characteristics (age, gender, and race). The dependent variable takes the value of 1 if student *i* and student *j* both listed each other as a friend. *dGPA* indicates a difference in average GPA within a pair of students. *B* indicates a Black student, and *W* indicates a White student. *BW* is an indicator that the sender is a Black student and the receiver is a White student. “Mean” refers to the average probability of forming friendships. “Controls” indicate the controlling demographic variables including gender and age of senders and receivers, difference in age within pairs, and school fixed effects. “Individual FE” indicates that individual fixed effects of senders and receivers are controlled for. Standard errors are clustered at the school level. *** $p < .01$, ** $p < .05$, * $p < .1$

Table B.5: Friendship Formation by Race and Achievement: Mutual Friendship (Either)

	Dependent variable : <i>i</i> nominating <i>j</i> as a friend (%)				
	(1)	(2)	(3)	(4)	(5)
<i>dGPA</i>	-0.282*** (0.028)	-0.251*** (0.023)	-0.319*** (0.037)	-0.284*** (0.032)	-0.388*** (0.041)
<i>dGPA</i> × <i>BB</i>			0.058 (0.067)	0.012 (0.069)	0.041 (0.074)
<i>dGPA</i> × <i>BW</i>			0.270*** (0.036)	0.229*** (0.033)	0.289*** (0.038)
<i>dGPA</i> × <i>WB</i>			0.270*** (0.036)	0.229*** (0.033)	0.289*** (0.038)
<i>BB</i>			0.334 (0.267)	0.475* (0.275)	3.574*** (0.398)
<i>BW</i>			-0.926*** (0.122)	-0.935*** (0.093)	0.000 (0.000)
<i>WB</i>			-0.926*** (0.122)	-0.935*** (0.093)	0.000 (0.000)
Constant	1.060*** (0.094)	1.299*** (0.147)	1.168*** (0.125)	1.546*** (0.200)	1.503*** (0.080)
Observations	36,183,256	36,183,256	36,183,256	36,183,256	36,183,256
Mean	0.817	0.817	0.817	0.817	0.817
Controls		X		X	X
Individual FE					X

Notes. The table presents coefficients and standard errors from estimation of equation (1). Units of observation are directed dyads, based on the non-Hispanic Black and non-Hispanic White students who reported GPAs and individual characteristics (age, gender, and race). The dependent variable takes on the value of 1 if at least one of students *i* and *j* listed another student as a friend. *dGPA* indicates a difference in average GPA within a pair of students. *B* indicates a Black student, and *W* indicates a White student. *BW* is an indicator that the sender is a Black student and the receiver is a White student. “Mean” refers to the average probability of forming friendships. “Controls” indicate the controlling demographic variables including gender and age of senders and receivers, difference in age within pairs, and school fixed effects. “Individual FE” indicates that individual fixed effects of senders and receivers are controlled for. Standard errors are clustered at the school level. *** $p < .01$, ** $p < .05$, * $p < .1$

B.4 Asymmetric GPA Measures

In Appendix Table B.6, we further examine whether the GPA gap matters differently when students direct another with a higher or lower GPA. The variable *ReceivingHighGPA* indicates that the sender student (*i*) has a lower GPA than the receiver student (*j*). We find that the relationship between the GPA gap and friendship formation is weaker when the receiver has a higher GPA ($ReceiverHighGPA \times dGPA$). This suggests that the difference in GPA matters differently when a student directs another student with a higher or lower GPA. However, we do not find a significant difference between White-White pairs and Black-Black pairs, regardless of the direction from lower GPA to higher GPA or higher GPA to lower GPA ($ReceiverHighGPA \times dGPA \times BB$ and $dGPA \times BB$).

Table B.6: Friendship Formation by Race and Achievement: Asymmetric GPA

	Dependent variable : <i>i</i> nominating <i>j</i> as a friend (%)	
	(1)	(2)
<i>ReceiverHighGPA</i> $\times$ <i>dGPA</i>	0.032*** (0.006)	0.132*** (0.016)
<i>ReceiverHighGPA</i> $\times$ <i>dGPA</i> $\times$ <i>BB</i>	-0.018 (0.021)	-0.015 (0.020)
<i>ReceiverHighGPA</i> $\times$ <i>dGPA</i> $\times$ <i>BW</i>	-0.040*** (0.009)	-0.046*** (0.014)
<i>ReceiverHighGPA</i> $\times$ <i>dGPA</i> $\times$ <i>WB</i>	0.003 (0.012)	0.006 (0.014)
<i>ReceiverHighGPA</i> $\times$ <i>BB</i>	0.033 (0.025)	0.029 (0.023)
<i>ReceiverHighGPA</i> $\times$ <i>BW</i>	0.021** (0.008)	0.017* (0.010)
<i>ReceiverHighGPA</i> $\times$ <i>WB</i>	0.024** (0.011)	0.031*** (0.011)
<i>ReceiverHighGPA</i>	-0.030*** (0.007)	-0.019*** (0.005)
<i>dGPA</i>	-0.229*** (0.027)	-0.253*** (0.028)
<i>dGPA</i> $\times$ <i>BB</i>	0.054 (0.044)	0.022 (0.045)
<i>dGPA</i> $\times$ <i>BW</i>	0.209*** (0.026)	0.187*** (0.024)
<i>dGPA</i> $\times$ <i>WB</i>	0.186*** (0.026)	0.158*** (0.025)
<i>BB</i>	0.153 (0.164)	0.240 (0.169)
<i>BW</i>	-0.631*** (0.079)	-0.636*** (0.061)
<i>WB</i>	-0.608*** (0.079)	-0.616*** (0.061)
Constant	0.762*** (0.081)	0.868*** (0.129)
Observations	36,519,842	36,183,256
Mean	0.5	0.5
Controls		X

Notes. The table presents coefficients and standard errors from estimation of equation (1). Units of observation are directed dyads, based on the non-Hispanic Black and non-Hispanic White students who reported GPAs and individual characteristics (age, gender, and race). The dependent variable takes the value of 1 if student *i* nominates student *j* as a friend. *ReceiverHighGPA* takes the value of 1 if the nominated student *j* has higher GPA than the student *i*. *dGPA* indicates a difference in average GPA within a pair of students. *B* indicates a Black student, and *W* indicates a White student. *BW* is an indicator that the sender is a Black student and the receiver is a White student. Mean is the average probability of forming friendships. Controls indicate the controlling demographic variables including gender and age of senders and receivers, difference in age within pairs, and school fixed effects. Standard errors are clustered at the school level. *** $p < .01$, ** $p < .05$, * $p < .1$

B.5 Semiparametric GPA Measures

We explore whether low and high GPA students exhibit different patterns when they form friendships with low and high GPA peers. We divide students into two GPA groups: low (average GPA 1-3) and high (average GPA 3-4). Then, we examine the difference across races when a student i with a low (or high) GPA forms a friendship with a student j with a low (or high) GPA. The results are presented in Appendix Table B.7. First, we observe a higher probability of forming friendships within the same GPA groups than across GPA groups. Particularly, stronger homophily exists among high GPA students. Regardless of whether a student with a low (or high) GPA forms friendships with a student with a low (or high) GPA, however, the GPA gap matters less or similarly to Black-Black pairs compared to White-White pairs. Additionally, we further explore the differences using three levels of GPA: low (average GPA 1-2.5), middle (average GPA 2.5-3.25), and high (average GPA 3.25-4), as shown in Appendix Table B.8. Once again, we find that the GPA gap matters similarly or less on friendships for Black-Black pairs compared to White-White pairs.

Table B.7: Friendship Formation by Race and Achievement: Semiparametric Low-High

	Dependent variable : <i>i</i> nominating <i>j</i> as a friend (%)	
	(1)	(2)
<i>GPA_i : low</i> × <i>GPA_j : high</i> × <i>BW</i>	0.060*** (0.022)	0.053** (0.021)
<i>GPA_i : low</i> × <i>GPA_j : high</i> × <i>WB</i>	0.113*** (0.025)	0.104*** (0.021)
<i>GPA_i : low</i> × <i>GPA_j : high</i> × <i>BB</i>	0.076 (0.047)	0.069** (0.031)
<i>GPA_i : high</i> × <i>GPA_j : low</i> × <i>BW</i>	0.115*** (0.024)	0.106*** (0.021)
<i>GPA_i : high</i> × <i>GPA_j : low</i> × <i>WB</i>	0.060** (0.023)	0.053** (0.023)
<i>GPA_i : high</i> × <i>GPA_j : low</i> × <i>BB</i>	0.040 (0.045)	0.032 (0.027)
<i>GPA_i : high</i> × <i>GPA_j : high</i> × <i>BW</i>	-0.214*** (0.075)	-0.221*** (0.055)
<i>GPA_i : high</i> × <i>GPA_j : high</i> × <i>WB</i>	-0.209*** (0.075)	-0.216*** (0.053)
<i>GPA_i : high</i> × <i>GPA_j : high</i> × <i>BB</i>	0.174 (0.244)	0.126 (0.192)
<i>GPA_i : low</i> × <i>GPA_j : high</i>	-0.074*** (0.021)	-0.066*** (0.009)
<i>GPA_i : high</i> × <i>GPA_j : low</i>	-0.087*** (0.023)	-0.089*** (0.012)
<i>GPA_i : high</i> × <i>GPA_j : high</i>	0.298*** (0.075)	0.283*** (0.041)
<i>BB</i>	0.214* (0.114)	0.237* (0.129)
<i>BW</i>	-0.452*** (0.057)	-0.499*** (0.052)
<i>WB</i>	-0.429*** (0.057)	-0.473*** (0.052)
Constant	0.548*** (0.058)	0.846*** (0.124)
Observations	36,183,256	36,183,256
Mean	0.5	0.5
Controls	X	X

Notes. The table presents coefficients and standard errors from estimation of equation (1), with GPA difference defined in intervals (high and low) rather than continuously. Units of observation are directed dyads, based on the non-Hispanic Black and non-Hispanic White students who reported GPAs and individual characteristics (age, gender, and race). The dependent variable takes the value of 1 if student *i* nominates student *j* as a friend. *B* indicates a Black student, and *W* indicates a White student. *BW* is an indicator that the sender is a Black student and the receiver is a White student. “Mean” refers to the average probability of forming friendships. “Controls” indicate the controlling demographic variables including gender and age of senders and receivers, difference in age within pairs, and school fixed effects. Standard errors are clustered at the school level. *** $p < .01$, ** $p < .05$, * $p < .1$

Table B.8: Forming Friends and Difference in GPA: Semiparametric Low-Middle-High

	Dependent variable : <i>i</i> nominating <i>j</i> as a friend (%)	
	(1)	(2)
$GPA_i : low \times GPA_j : middle \times BW$	0.086*** (0.018)	0.093*** (0.024)
$GPA_i : low \times GPA_j : middle \times WB$	0.093*** (0.017)	0.081*** (0.020)
$GPA_i : low \times GPA_j : middle \times BB$	0.046 (0.044)	0.057 (0.039)
$GPA_i : low \times GPA_j : high \times BW$	0.187*** (0.032)	0.167*** (0.034)
$GPA_i : low \times GPA_j : high \times WB$	0.241*** (0.035)	0.217*** (0.034)
$GPA_i : low \times GPA_j : high \times BB$	0.107** (0.052)	0.099** (0.039)
$GPA_i : middle \times GPA_j : low \times BW$	0.104*** (0.018)	0.090*** (0.024)
$GPA_i : middle \times GPA_j : low \times WB$	0.080*** (0.020)	0.086*** (0.026)
$GPA_i : middle \times GPA_j : low \times BB$	0.076* (0.041)	0.084*** (0.031)
$GPA_i : middle \times GPA_j : middle \times BW$	0.003 (0.031)	0.021 (0.039)
$GPA_i : middle \times GPA_j : middle \times WB$	-0.001 (0.033)	0.018 (0.038)
$GPA_i : middle \times GPA_j : middle \times BB$	0.076 (0.097)	0.094 (0.078)
$GPA_i : middle \times GPA_j : high \times BW$	0.012 (0.047)	-0.008 (0.043)
$GPA_i : middle \times GPA_j : high \times WB$	0.032 (0.048)	0.041 (0.042)
$GPA_i : middle \times GPA_j : high \times BB$	0.230 (0.173)	0.204 (0.139)
$GPA_i : high \times GPA_j : low \times BW$	0.248*** (0.033)	0.224*** (0.033)
$GPA_i : high \times GPA_j : low \times WB$	0.179*** (0.033)	0.160*** (0.035)
$GPA_i : high \times GPA_j : low \times BB$	0.090* (0.051)	0.083** (0.038)
$GPA_i : high \times GPA_j : middle \times BW$	0.068 (0.047)	0.078* (0.040)
$GPA_i : high \times GPA_j : middle \times WB$	0.019 (0.047)	0.001 (0.043)
$GPA_i : high \times GPA_j : middle \times BB$	0.153 (0.151)	0.131 (0.117)

Table B.8: Friendship Formation by Race and Achievement: Semiparametric Low-Middle-High (Continued)

	Dependent variable : <i>i</i> nominating <i>j</i> as a friend (%)	
	(1)	(2)
<i>GPA_i : high</i> × <i>GPA_j : high</i> × <i>BW</i>	-0.207** (0.091)	-0.229*** (0.070)
<i>GPA_i : high</i> × <i>GPA_j : high</i> × <i>WB</i>	-0.209** (0.090)	-0.229*** (0.066)
<i>GPA_i : high</i> × <i>GPA_j : high</i> × <i>BB</i>	0.233 (0.305)	0.165 (0.245)
<i>GPA_i : low</i> × <i>GPA_j : middle</i>	-0.085*** (0.015)	-0.067*** (0.009)
<i>GPA_i : low</i> × <i>GPA_j : high</i>	-0.211*** (0.030)	-0.178*** (0.020)
<i>GPA_i : middle</i> × <i>GPA_j : low</i>	-0.098*** (0.017)	-0.087*** (0.011)
<i>GPA_i : middle</i> × <i>GPA_j : middle</i>	0.018 (0.028)	0.031** (0.015)
<i>GPA_i : middle</i> × <i>GPA_j : high</i>	0.013 (0.045)	0.037** (0.016)
<i>GPA_i : high</i> × <i>GPA_j : low</i>	-0.229*** (0.032)	-0.210*** (0.024)
<i>GPA_i : high</i> × <i>GPA_j : middle</i>	-0.005 (0.044)	0.014 (0.015)
<i>GPA_i : high</i> × <i>GPA_j : high</i>	0.307*** (0.090)	0.318*** (0.050)
<i>BB</i>	0.173* (0.101)	0.197* (0.118)
<i>BW</i>	-0.500*** (0.061)	-0.544*** (0.055)
<i>WB</i>	-0.469*** (0.061)	-0.511*** (0.055)
Constant	0.592*** (0.062)	0.877*** (0.124)
Observations	36,183,256	36,183,256
Mean	0.5	0.5
Controls		X

Notes. The table presents coefficients and standard errors from estimation of equation (1), with GPA difference defined in intervals (high, middle, and low) rather than continuously. Units of observation are directed dyads, based on the non-Hispanic Black and non-Hispanic White students who reported GPAs and individual characteristics (age, gender, and race). The dependent variable takes the value of 1 if student *i* nominates student *j* as a friend. *B* indicates a Black student, and *W* indicates a White student. *BW* is an indicator that the sender is a Black student and the receiver is a White student. “Mean” refers to the average probability of forming friendships. “Controls” indicate the controlling demographic variables including gender and age of senders and receivers, difference in age within pairs, and school fixed effects. Standard errors are clustered at the school level. *** $p < .01$, ** $p < .05$, * $p < .1$

B.6 Including Hispanic Students

We examine whether including Hispanic students affects our results by expanding the analysis sample to include Hispanic students and adding corresponding interaction terms to regression equation (1). The added variables are: B_iH_j , W_iH_j , H_iH_j , H_iW_j , H_iB_j , and their interactions with GPA differences: $d_{ij} \times B_iH_j$, $d_{ij} \times W_iH_j$, $d_{ij} \times H_iH_j$, $d_{ij} \times H_iW_j$, and $d_{ij} \times H_iB_j$. As shown in Appendix Table B.9, the inclusion of Hispanic students does not significantly alter our main findings. GPA differences remain similarly important for friendship formation among Black–Black and White–White pairs ($dGPA \times BB$) and are less influential for Black–White pairs ($dGPA \times BW$).

Table B.9: Friendship Formation by Race and Achievement: Including Hispanic Students

	Dependent variable : <i>i</i> nominating <i>j</i> as a friend (%)	
	(1)	(2)
<i>dGPA</i>	-0.216*** (0.025)	-0.195*** (0.022)
<i>dGPA</i> × <i>BB</i>	0.047 (0.044)	0.022 (0.045)
<i>dGPA</i> × <i>BW</i>	0.188*** (0.025)	0.163*** (0.023)
<i>dGPA</i> × <i>BH</i>	0.186*** (0.028)	0.158*** (0.026)
<i>dGPA</i> × <i>WB</i>	0.182*** (0.024)	0.160*** (0.023)
<i>dGPA</i> × <i>WH</i>	0.110*** (0.023)	0.104*** (0.020)
<i>dGPA</i> × <i>HH</i>	0.151*** (0.034)	0.117*** (0.034)
<i>dGPA</i> × <i>HW</i>	0.117*** (0.023)	0.107*** (0.019)
<i>dGPA</i> × <i>HB</i>	0.180*** (0.028)	0.152*** (0.025)
<i>BB</i>	0.168 (0.166)	0.234 (0.152)
<i>BW</i>	-0.624*** (0.078)	-0.612*** (0.059)
<i>BH</i>	-0.541*** (0.095)	-0.324*** (0.066)
<i>WB</i>	-0.597*** (0.078)	-0.587*** (0.058)
<i>WH</i>	-0.360*** (0.071)	-0.261*** (0.050)
<i>HH</i>	-0.427*** (0.132)	-0.060 (0.074)
<i>HW</i>	-0.383*** (0.070)	-0.281*** (0.048)
<i>HB</i>	-0.535*** (0.094)	-0.315*** (0.067)
Constant	0.752*** (0.080)	0.876*** (0.092)
Observations	53,008,910	53,008,910
Mean	0.433	0.433
Controls		X

Notes. The table presents coefficients and standard errors from estimation of equation (1). Units of observation are directed dyads, based on the non-Hispanic Black, non-Hispanic White, and Hispanic students who reported GPAs and individual characteristics (age, gender, and race). The dependent variable takes the value of 1 if student *i* nominates student *j* as a friend. *dGPA* indicates a difference in average GPA within a pair of students. *B*, *W*, and *H* indicate a Black, White, and Hispanic student, respectively. “Mean” refers to the average probability of forming friendships. “Controls” indicate the controlling demographic variables including gender and age of senders and receivers, difference in age within pairs, and school fixed effects. Standard errors are clustered at the school level.

*** $p < .01$, ** $p < .05$, * $p < .1$

References for Online Appendix

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